

# Libertinus and diacritics

## Anchor model

Libertinus has an anchor model where the base anchors are set to locations where the marks would be placed relative to the base. See the blue dots in the left panel of Figure 1; from top to bottom, the above-anchor is the blue dot above the base glyph, the overlay-anchor is blue dot inside the base glyph outline, and the below-anchor is the blue dot below the base glyph. The mark anchor is generally near the optical center of the mark glyph outline. See the red square in the middle panel of Figure 1; the mark's below-anchor is at the optical center of the mark. The base and mark are combined in the right panel of Figure 1 by superimposing their below-anchors.

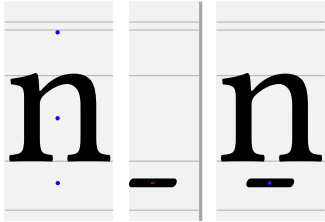


Figure 1: Libertinus n and macron with anchors.

Other fonts use a different geometric model for anchors. In Noto, the base's below-anchor is at the baseline, the overlay-anchor is inside the glyph, and the above-anchor is at the x-height. See the blue dots in the left panel of Figure 2; from the bottom to the top, the mark's below-anchor is the blue dot on the baseline, the overlay-anchor is blue dot inside the base glyph outline, and the above-anchor is the blue dot at the x-height. In Noto, the mark's below-anchor is at the origin (0, 0), and the mark glyph outline is drawn below the origin. In the middle panel of Figure 2, the mark's below-anchor is the red square at the origin. The base and mark are combined in the right panel of Figure 2 by superimposing their below-anchors.

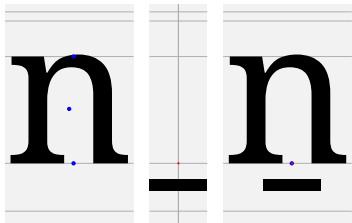


Figure 2: Noto n and macron with anchors.

## Font metrics per glyph

See above- and below-anchor coordinates for A–Z, a–z, and IPA letters for Libertinus serif fonts (regular, italic, semibold, semibold italic) in four tables (Tables 1, 2, 3, 4, 5, 6). Recall that Language Science Press uses semibold Libertinus fonts, e.g. `LibertinusSerif-Semibold.otf`, for all boldface style.

### Table row headings

hex: Unicode character codepoint in 4-hex.

reg, it, sb, si: regular, italic, semibold, semibold italic. In that row, the character is printed alone, with combining dot above, and with combining macron below, to show any anchor positioning. If no above- or below-anchor exists for that character, the character and combining mark are positioned by HarfBuzz fallback shaping.

$ax, ay$ :  $X$  and  $Y$  coordinates of the above-anchor.

$bx, by$ :  $X$  and  $Y$  coordinates of the below-anchor.

The next few rows use the bounding box of the glyph, because the midpoint of the width of the bounding box is a naive candidate for  $ax$  and  $bx$ , and the bounding box coordinates are easy to determine.

$\square xm$ : midpoint of the bounding box  $X$  coordinates,  $(X_{\min} + X_{\max})/2$ , where the bounding box is characterized by  $(X_{\min}, Y_{\min})$  and  $(X_{\max}, Y_{\max})$ .

$ax\delta$ :  $X$  coordinate offset of above-anchor from midpoint of bounding box,  $ax - \square xm$ .

$ax\%$ : normalized offset,  $ax\delta$  divided by bounding box width,  $ax\delta / (X_{\max} - X_{\min})$ .

$bx\delta$ :  $X$  coordinate offset of below-anchor from midpoint of bounding box,  $bx - \square xm$ .

$bx\%$ : normalized offset,  $bx\delta$  divided by bounding box width,  $bx\delta / (X_{\max} - X_{\min})$ .

There are better ways to define the geometric horizontal center of a glyph than the bounding box midpoint, depending on the outline at the top or bottom of the glyph.

## Vertical aspects, anchor $Y$ coordinates and reference, and clearance

A font has at least five vertical aspects where glyph outlines live inside: descender height, baseline, x-height, ascender height, capital height. The values or ranges of these  $Y$  coordinates are in the first five rows of Table 9. These vertical aspects are the gray horizontal lines of Figure 1. These vertical aspects are displayed in a FontForge glyph outline window as black horizontal guidelines with red or blue shaded bands for ranges. I denote these as  $yra, yrc, yrx, yr0, yrd$  because they are  $Y$  values, and r indicates a reference value. a, c, x, 0, d indicate the five vertical aspects.

In the anchor model for Libertinus, the  $Y$  coordinate of an above-anchor ( $ay$ ) or a below-anchor ( $by$ ) is at a stable value depending on where the glyph outline lives relative to a vertical aspect. See the rows for  $ay$  and  $by$  in Tables 1 and 2 and find the common values according to the glyph's vertical aspects. There is no set name in font design for these above- or below-anchor  $Y$  coordinate reference values by ascender aspect, so I will denote them  $ayra, ayrc, ayrx, byr0, byrd$ . These are the middle five rows of Table 9. For example, all glyphs that sit on the baseline have below-anchor  $Y$  coordinate  $by = -110$ , and thus  $byr0 = -110$ . All capitals A–Z have  $ay = 850$ , and thus  $ayrc = 850$ . Any glyph that descends to the descender value  $yrd = -232$  or descender range  $[-238, -227]$  should have  $by = byrd = -319$ .

The clearance is defined as the difference between the anchor reference value and the vertical aspect  $Y$  value. I will denote these clearances as  $ay\delta a, ay\delta c, ay\delta x, by\delta 0, by\delta d$ . So,  $ay\delta a = ayra - yra$  etc. For example, capital letters A–Z have a capital height  $yrc = 645$ , and all capital letters A–Z have an above-anchor  $Y$  coordinate at  $ayrc = 850$ . So the above-anchor clearance for capitals is  $ay\delta c = ayrc - yec = 850 - 645 = 205$ . Because of the weird anchor model for Libertinus, this clearance value is not the exact vertical distance between a base and a mark, but this clearance value is still useful.

A glyph may not be a full ascender or a full descender, and its bounding box may not live closely within two vertical aspects. In such a case, the anchor  $Y$  coordinate is set by looking for the nearest, or most suitable, vertical aspect, and considering the clearance of that vertical aspect relative to the bounding box of the glyph.

Here is the heuristic algorithm for calculating the  $Y$  coordinate of the above- or below-anchor ( $ay$  or  $by$ ) of a glyph that is not a full ascender or descender.

Table 1: Font metrics for A–Z

| hex | 0041  | 0042  | 0043  | 0044  | 0045  | 0046  | 0047  | 0048  | 0049  | 004A  | 004B  | 004C  | 004D  | 004E  | 004F  | 0050  | 0051  | 0052  | 0053  | 0054  | 0055  | 0056  | 0057  | 0058  | 0059  | 005A  |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| reg | AAA   | BBB   | CCC   | DDD   | EEE   | FFF   | GGG   | HHH   | III   | JJJ   | KKK   | LLL   | MMM   | NNN   | OOO   | PPP   | QQQ   | RRR   | SSS   | TTT   | UUU   | VVV   | WWW   | XXX   | YYY   | ZZZ   |
| ax  | 354   | 267   | 340   | 306   | 281   | 284   | 365   | 361   | 150   | 182   | 336   | 151   | 431   | 346   | 335   | 277   | 348   | 269   | 232   | 307   | 346   | 336   | 450   | 332   | 295   | 320   |
| ay  | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   |
| bx  | 308   | 268   | 363   | 309   | 287   | 209   | 367   | 362   | 147   | 109   | 342   | 262   | 393   | 348   | 350   | 240   |       | 258   | 234   | 300   | 328   | 318   | 479   | 307   | 295   | 287   |
| by  | -110  | -110  | -110  | -110  | -110  | -110  | -110  | -110  | -110  | -319  | -110  | -110  | -110  | -110  | -110  | -110  |       | -110  | -110  | -110  | -110  | -110  | -110  | -110  | -110  | -110  |
| □xm | 346   | 280   | 324   | 336   | 271   | 244   | 351   | 362   | 148   | 120   | 323   | 260   | 418   | 353   | 351   | 259   | 359   | 301   | 235   | 297   | 331   | 325   | 474   | 323   | 287   | 309   |
| axδ | 8     | -13   | 16    | -30   | 10    | 39    | 13    | -1    | 2     | 62    | 13    | -109  | 13    | -7    | -16   | 17    | -11   | -32   | -3    | 10    | 15    | 10    | -24   | 9     | 7     | 10    |
| ax% | 0.01  | -0.03 | 0.03  | -0.05 | 0.02  | 0.09  | 0.02  | -0.00 | 0.01  | 0.16  | 0.02  | -0.22 | 0.02  | -0.01 | -0.03 | 0.04  | -0.02 | -0.06 | -0.01 | 0.02  | 0.02  | 0.02  | -0.03 | 0.01  | 0.01  | 0.02  |
| bxδ | -38   | -12   | 39    | -27   | 16    | -35   | 15    | 0     | -1    | -11   | 19    | 1     | -25   | -5    | -1    | -19   |       | -43   | -1    | 3     | -3    | -7    | 5     | -16   | 7     | -22   |
| bx% | -0.06 | -0.02 | 0.07  | -0.04 | 0.03  | -0.08 | 0.02  | 0.00  | -0.00 | -0.03 | 0.03  | 0.00  | -0.03 | -0.01 | -0.00 | -0.04 |       | -0.07 | -0.00 | 0.01  | -0.00 | -0.01 | 0.01  | -0.02 | 0.01  | -0.04 |
| it  | AAA   | BBB   | CCC   | DDD   | EEE   | FFF   | GGG   | HHH   | III   | JJJ   | KKK   | LLL   | MMM   | NNN   | OOO   | PPP   | QQQ   | RRR   | SSS   | TTT   | UUU   | VVV   | WWW   | XXX   | YYY   | ZZZ   |
| ax  | 505   | 447   | 500   | 396   | 413   | 406   | 513   | 481   | 316   | 392   | 466   | 389   | 557   | 476   | 515   | 426   | 498   | 419   | 402   | 457   | 486   | 481   | 610   | 482   | 435   | 450   |
| ay  | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   |
| bx  | 308   | 238   | 333   | 279   | 269   | 131   | 345   | 320   | 127   | 109   | 292   | 252   | 363   | 328   | 310   | 140   |       | 248   | 204   | 277   | 308   | 263   | 439   | 287   | 270   | 267   |
| by  | -108  | -108  | -110  | -107  | -107  | -111  | -110  | -107  | -109  | -319  | -110  | -110  | -110  | -109  | -113  | -110  |       | -111  | -111  | -109  | -109  | -110  | -107  | -107  | -109  | -107  |
| □xm | 334   | 289   | 379   | 355   | 282   | 282   | 395   | 398   | 203   | 263   | 370   | 268   | 445   | 400   | 396   | 295   | 396   | 295   | 272   | 383   | 441   | 418   | 587   | 367   | 393   | 357   |
| axδ | 170   | 158   | 120   | 41    | 131   | 123   | 118   | 83    | 113   | 128   | 95    | 120   | 112   | 76    | 118   | 131   | 101   | 124   | 130   | 73    | 45    | 63    | 23    | 114   | 42    | 93    |
| ax% | 0.26  | 0.29  | 0.21  | 0.06  | 0.24  | 0.23  | 0.19  | 0.11  | 0.31  | 0.28  | 0.14  | 0.24  | 0.13  | 0.10  | 0.19  | 0.23  | 0.16  | 0.22  | 0.29  | 0.13  | 0.07  | 0.10  | 0.03  | 0.16  | 0.08  | 0.15  |
| bxδ | -26   | -51   | -46   | -76   | -13   | -151  | -50   | -78   | -76   | -154  | -78   | -16   | -82   | -72   | -86   | -155  |       | -47   | -68   | -106  | -133  | -155  | -148  | -80   | -123  | -90   |
| bx% | -0.04 | -0.09 | -0.08 | -0.11 | -0.02 | -0.28 | -0.08 | -0.10 | -0.21 | -0.33 | -0.11 | -0.03 | -0.10 | -0.09 | -0.14 | -0.28 |       | -0.08 | -0.15 | -0.20 | -0.21 | -0.25 | -0.16 | -0.11 | -0.23 | -0.14 |
| sb  | AAA   | BBB   | CCC   | DDD   | EEE   | FFF   | GGG   | HHH   | III   | JJJ   | KKK   | LLL   | MMM   | NNN   | OOO   | PPP   | QQQ   | RRR   | SSS   | TTT   | UUU   | VVV   | WWW   | XXX   | YYY   | ZZZ   |
| ax  |       |       |       |       |       |       |       |       |       | 203   |       |       |       |       |       | 363   | 363   | 304   | 252   | 298   |       | 331   | 493   | 364   | 316   | 321   |
| ay  |       |       |       |       |       |       |       |       |       | 806   |       |       |       |       |       | 802   | 802   | 808   | 802   | 802   |       | 805   | 805   | 804   | 805   | 803   |
| bx  |       |       |       |       |       |       |       | 362   |       | 109   |       |       |       |       |       |       |       | 292   | 234   | 301   |       | 303   | 470   | 318   | 316   | 308   |
| by  |       |       |       |       |       |       |       | -110  |       | -319  |       |       |       |       |       |       |       | -111  | -111  | -107  |       | -108  | -107  | -107  | -109  | -107  |
| □xm | 349   | 299   | 332   | 351   | 283   | 268   | 368   | 362   | 160   | 131   | 340   | 267   | 427   | 354   | 365   | 277   | 365   | 320   | 237   | 299   | 339   | 318   | 489   | 354   | 308   | 309   |
| axδ |       |       |       |       |       |       |       |       |       | 72    |       |       |       |       |       | -2    | -2    | -16   | 14    | -1    |       | 12    | 4     | 9     | 7     | 12    |
| ax% |       |       |       |       |       |       |       |       |       | 0.17  |       |       |       |       |       | -0.00 | -0.00 | -0.03 | 0.04  | -0.00 |       | 0.02  | 0.00  | 0.01  | 0.01  | 0.02  |
| bxδ |       |       |       |       |       |       | 0     |       |       | -22   |       |       |       |       |       |       |       | -28   | -3    | 2     |       | -15   | -19   | -36   | 7     | -1    |
| bx% |       |       |       |       |       |       | 0.00  |       |       | -0.05 |       |       |       |       |       |       |       | -0.05 | -0.01 | 0.00  |       | -0.02 | -0.02 | -0.05 | 0.01  | -0.00 |
| si  | AAA   | BBB   | CCC   | DDD   | EEE   | FFF   | GGG   | HHH   | III   | JJJ   | KKK   | LLL   | MMM   | NNN   | OOO   | PPP   | QQQ   | RRR   | SSS   | TTT   | UUU   | VVV   | WWW   | XXX   | YYY   | ZZZ   |
| ax  | 354   | 267   | 340   | 306   | 273   | 276   | 365   | 361   | 150   | 432   | 336   | 149   | 431   | 346   | 335   | 316   | 348   | 269   | 232   | 307   | 346   | 336   | 450   | 332   | 295   | 320   |
| ay  | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   | 850   |
| bx  | 308   | 268   | 363   | 309   | 279   | 201   | 367   | 365   | 147   | 109   | 342   | 260   | 393   | 348   | 350   | 240   |       | 258   | 234   | 300   | 328   | 318   | 479   | 307   | 295   | 287   |
| by  | -110  | -110  | -110  | -110  | -110  | -110  | -110  | -110  | -110  | -319  | -110  | -110  | -110  | -110  | -110  | -110  |       | -110  | -110  | -110  | -110  | -110  | -110  | -110  | -110  | -110  |
| □xm | 364   | 298   | 378   | 367   | 305   | 304   | 405   | 445   | 244   | 288   | 421   | 312   | 449   | 452   | 435   | 307   | 435   | 309   | 263   | 381   | 470   | 417   | 581   | 387   | 407   | 350   |
| axδ | -10   | -31   | -38   | -61   | -32   | -28   | -40   | -84   | -94   | 144   | -85   | -163  | -18   | -106  | -100  | 8     | -87   | -40   | -31   | -74   | -124  | -81   | -131  | -55   | -112  | -30   |
| ax% | -0.01 | -0.05 | -0.06 | -0.09 | -0.06 | -0.05 | -0.06 | -0.11 | -0.23 | 0.30  | -0.12 | -0.31 | -0.02 | -0.13 | -0.15 | 0.01  | -0.13 | -0.07 | -0.07 | -0.13 | -0.18 | -0.13 | -0.14 | -0.07 | -0.19 | -0.05 |
| bxδ | -56   | -30   | -15   | -88   | -26   | -103  | -38   | -80   | -97   | -179  | -79   | -52   | -56   | -104  | -85   | -67   |       | -51   | -29   | -81   | -142  | -99   | -102  | -80   | -112  | -63   |
| bx% | -0.08 | -0.05 | -0.03 | -0.13 | -0.05 | -0.18 | -0.06 | -0.10 | -0.23 | -0.37 | -0.11 | -0.10 | -0.07 | -0.13 | -0.13 | -0.12 |       | -0.09 | -0.07 | -0.14 | -0.21 | -0.16 | -0.11 | -0.11 | -0.19 | -0.10 |

Table 2: Font metrics for a–z

| hex | 0061      | 0062      | 0063      | 0064      | 0065      | 0066      | 0067      | 0068      | 0069      | 006A      | 006B      | 006C      | 006D      | 006E      | 006F      | 0070      | 0071      | 0072      | 0073      | 0074      | 0075      | 0076      | 0077      | 0078      | 0079      | 007A      |
|-----|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| reg | <i>aa</i> | <i>bb</i> | <i>cc</i> | <i>dd</i> | <i>ee</i> | <i>ff</i> | <i>gg</i> | <i>hh</i> | <i>ii</i> | <i>jj</i> | <i>kk</i> | <i>ll</i> | <i>mm</i> | <i>nn</i> | <i>oo</i> | <i>pp</i> | <i>qq</i> | <i>rr</i> | <i>ss</i> | <i>tt</i> | <i>uu</i> | <i>vv</i> | <i>ww</i> | <i>xx</i> | <i>yy</i> | <i>zz</i> |
| ax  | 209       |           | 242       | 218       | 243       |           | 231       | 299       | 137       |           | 306       | 135       | 392       | 264       | 238       | 271       | 238       | 191       | 206       |           | 255       | 256       | 345       | 256       | 262       | 220       |
| ay  | 645       |           | 645       | 645       | 645       |           | 645       | 645       | 795       |           | 784       | 885       | 645       | 646       | 645       | 646       | 645       | 646       | 645       |           | 645       | 646       | 646       | 646       | 645       | 646       |
| bx  | 196       | 234       | 236       | 221       | 234       | 132       | 229       | 272       | 136       | 109       | 290       | 135       | 408       | 265       | 243       | 179       | 348       | 135       | 196       | 163       | 239       | 241       | 352       | 224       | 242       | 207       |
| by  | -110      | -110      | -110      | -110      | -110      | -110      | -319      | -110      | -110      | -319      | -214      | -110      | -110      | -110      | -110      | -319      | -319      | -110      | -110      | -110      | -110      | -110      | -110      | -110      | -319      | -110      |
| □xm | 245       | 231       | 217       | 270       | 222       | 208       | 256       | 270       | 142       | 75        | 261       | 134       | 401       | 275       | 252       | 242       | 268       | 190       | 199       | 163       | 271       | 249       | 373       | 245       | 260       | 220       |
| axδ | -36       |           | 24        | -52       | 21        |           | -25       | 28        | -5        |           | 45        | 1         | -9        | -11       | -14       | 28        | -30       | 1         | 7         |           | -16       | 7         | -28       | 11        | 2         | 0         |
| ax% | -0.09     |           | 0.07      | -0.11     | 0.06      |           | -0.06     | 0.06      | -0.02     |           | 0.09      | 0.00      | -0.01     | -0.02     | -0.03     | 0.06      | -0.07     | 0.00      | 0.02      |           | -0.03     | 0.01      | -0.04     | 0.02      | 0.00      | -0.00     |
| bxδ | -49       | 2         | 18        | -49       | 12        | -76       | -27       | 1         | -6        | 33        | 29        | 1         | 6         | -10       | -9        | -63       | 79        | -55       | -3        | 0         | -32       | -8        | -21       | -21       | -18       | -13       |
| bx% | -0.12     | 0.01      | 0.05      | -0.11     | 0.03      | -0.21     | -0.06     | 0.00      | -0.03     | 0.14      | 0.06      | 0.00      | 0.01      | -0.02     | -0.02     | -0.13     | 0.17      | -0.16     | -0.01     | 0.00      | -0.06     | -0.02     | -0.03     | -0.05     | -0.04     | -0.04     |
| it  | <i>aa</i> | <i>bb</i> | <i>cc</i> | <i>dd</i> | <i>ee</i> | <i>ff</i> | <i>gg</i> | <i>hh</i> | <i>ii</i> | <i>jj</i> | <i>kk</i> | <i>ll</i> | <i>mm</i> | <i>nn</i> | <i>oo</i> | <i>pp</i> | <i>qq</i> | <i>rr</i> | <i>ss</i> | <i>tt</i> | <i>uu</i> | <i>vv</i> | <i>ww</i> | <i>xx</i> | <i>yy</i> | <i>zz</i> |
| ax  | 341       |           | 343       | 218       | 332       |           | 360       | 409       |           |           | 405       | 292       | 522       | 394       | 359       | 411       | 369       | 312       | 327       |           | 386       |           | 348       | 366       | 367       | 360       |
| ay  | 647       |           | 645       | 645       | 645       |           | 645       | 645       |           |           | 646       | 890       | 645       | 646       | 645       | 646       | 645       | 646       | 645       |           | 646       |           | 646       | 646       | 644       | 646       |
| bx  | 174       | 214       | 163       | 221       | 158       | 222       | 197       | 229       | 136       | 0         | 223       | 115       | 382       | 235       | 186       | 248       | 169       | 113       | 169       | 133       | 226       |           | 355       | 224       | 368       | 187       |
| by  | -108      | -108      | -113      | -107      | -108      | -111      | -319      | -111      | -107      | -319      | -107      | -107      | -105      | -111      | -107      | -110      | -110      | -110      | -111      | -107      | -111      |           | -110      | -110      | -107      | -111      |
| □xm | 285       | 295       | 253       | 314       | 253       | 222       | 261       | 308       | 202       | 112       | 298       | 202       | 449       | 304       | 271       | 245       | 294       | 259       | 218       | 232       | 311       | 300       | 408       | 285       | 299       | 260       |
| axδ | 56        |           | 90        | -96       | 78        |           | 98        | 100       |           |           | 107       | 90        | 73        | 89        | 87        | 166       | 75        | 53        | 109       |           | 75        |           | -60       | 80        | 68        | 99        |
| ax% | 0.14      |           | 0.26      | -0.21     | 0.23      |           | 0.19      | 0.23      |           |           | 0.24      | 0.41      | 0.11      | 0.21      | 0.23      | 0.31      | 0.18      | 0.16      | 0.34      |           | 0.17      |           | -0.09     | 0.16      | 0.14      | 0.24      |
| bxδ | -111      | -81       | -90       | -93       | -95       | 0         | -64       | -79       | -66       | -112      | -75       | -87       | -67       | -69       | -85       | 3         | -125      | -146      | -49       | -99       | -85       |           | -53       | -61       | 69        | -73       |
| bx% | -0.27     | -0.21     | -0.26     | -0.20     | -0.28     | -0.00     | -0.13     | -0.18     | -0.35     | -0.29     | -0.17     | -0.40     | -0.10     | -0.16     | -0.22     | 0.01      | -0.29     | -0.43     | -0.15     | -0.38     | -0.19     |           | -0.08     | -0.12     | 0.14      | -0.18     |
| sb  | <i>aa</i> | <i>bb</i> | <i>cc</i> | <i>dd</i> | <i>ee</i> | <i>ff</i> | <i>gg</i> | <i>hh</i> | <i>ii</i> | <i>jj</i> | <i>kk</i> | <i>ll</i> | <i>mm</i> | <i>nn</i> | <i>oo</i> | <i>pp</i> | <i>qq</i> | <i>rr</i> | <i>ss</i> | <i>tt</i> | <i>uu</i> | <i>vv</i> | <i>ww</i> | <i>xx</i> | <i>yy</i> | <i>zz</i> |
| ax  |           |           |           |           |           | 231       |           |           | 137       |           |           | 142       | 396       | 264       |           |           |           |           |           |           | 264       |           |           |           |           |           |
| ay  |           |           |           |           |           | 645       |           |           | 795       |           |           | 885       | 645       | 646       |           |           |           |           |           |           | 646       |           |           |           |           |           |
| bx  |           |           |           |           |           | 229       | 277       | 136       | 109       |           |           | 135       | 396       | 265       |           |           |           |           |           | 163       | 248       |           |           |           |           |           |
| by  |           |           |           |           |           | -319      | -110      | -107      | -319      |           |           | -110      | -105      | -111      |           |           |           |           |           | -107      | -111      |           |           |           |           |           |
| □xm | 255       | 235       | 224       | 273       | 232       | 225       | 263       | 275       | 155       | 78        | 261       | 148       | 408       | 289       | 259       | 254       | 273       | 212       | 201       | 168       | 275       | 262       | 384       | 260       | 264       | 221       |
| axδ |           |           |           |           |           | -32       |           | -18       |           |           |           | -6        | -12       | -25       |           |           |           |           |           |           | -11       |           |           |           |           |           |
| ax% |           |           |           |           |           | -0.07     |           | -0.08     |           |           |           | -0.03     | -0.02     | -0.05     |           |           |           |           |           |           | -0.02     |           |           |           |           |           |
| bxδ |           |           |           |           |           | -34       | 2         | -19       | 31        |           |           | -13       | -12       | -24       |           |           |           |           |           | -5        | -27       |           |           |           |           |           |
| bx% |           |           |           |           |           | -0.08     | 0.00      | -0.08     | 0.11      |           |           | -0.05     | -0.02     | -0.05     |           |           |           |           |           |           | -0.02     | -0.05     |           |           |           |           |
| si  | <i>aa</i> | <i>bb</i> | <i>cc</i> | <i>dd</i> | <i>ee</i> | <i>ff</i> | <i>gg</i> | <i>hh</i> | <i>ii</i> | <i>jj</i> | <i>kk</i> | <i>ll</i> | <i>mm</i> | <i>nn</i> | <i>oo</i> | <i>pp</i> | <i>qq</i> | <i>rr</i> | <i>ss</i> | <i>tt</i> | <i>uu</i> | <i>vv</i> | <i>ww</i> | <i>xx</i> | <i>yy</i> | <i>zz</i> |
| ax  | 379       |           | 242       | 218       | 243       |           | 291       | 469       | 137       |           | 306       | 174       | 392       | 264       | 238       | 271       | 238       | 191       | 326       |           | 255       | 256       | 345       |           | 262       | 220       |
| ay  | 645       |           | 645       | 645       | 645       |           | 645       | 645       | 795       |           | 784       | 890       | 645       | 646       | 645       | 646       | 645       | 646       | 645       |           | 645       | 646       | 646       |           | 645       | 646       |
| bx  | 196       | 234       | 236       | 221       | 234       | 132       | 199       | 272       | 136       | 67        | 290       | 135       | 400       | 265       | 243       | 179       | 348       | 135       | 196       | 133       | 239       | 241       | 352       |           | 242       | 207       |
| by  | -110      | -110      | -110      | -110      | -110      | -110      | -319      | -110      | -110      | -319      | -214      | -110      | -110      | -110      | -110      | -319      | -319      | -110      | -110      | -107      | -110      | -110      | -110      |           | -319      | -110      |
| □xm | 317       | 303       | 277       | 336       | 277       | 219       | 270       | 338       | 221       | 132       | 339       | 226       | 473       | 344       | 294       | 257       | 313       | 295       | 245       | 259       | 324       | 311       | 418       | 297       | 282       | 267       |
| axδ | 61        |           | -35       | -118      | -34       |           | 21        | 130       | -84       |           | -33       | -52       | -81       | -80       | -56       | 13        | -75       | -104      | 80        |           | -69       | -55       | -73       |           | -20       | -47       |
| ax% | 0.13      |           | -0.09     | -0.22     | -0.09     |           | 0.04      | 0.26      | -0.38     |           | -0.07     | -0.20     | -0.11     | -0.16     | -0.13     | 0.02      | -0.16     | -0.25     | 0.22      |           | -0.14     | -0.13     | -0.12     |           | -0.04     | -0.11     |
| bxδ | -121      | -69       | -41       | -115      | -43       | -87       | -71       | -66       | -85       | -65       | -49       | -91       | -73       | -79       | -51       | -78       | 35        | -160      | -49       | -126      | -85       | -70       | -66       |           | -40       | -60       |
| bx% | -0.25     | -0.14     | -0.10     | -0.22     | -0.11     | -0.14     | -0.13     | -0.13     | -0.38     | -0.15     | -0.10     | -0.35     | -0.10     | -0.16     | -0.12     | -0.13     | 0.07      | -0.39     | -0.14     | -0.40     | -0.17     | -0.16     | -0.11     |           | -0.08     | -0.14     |

|     |      |      |       |      |      |       |      |       |      |      |      |      |      |       |      |      |      |      |      |       |      |      |      |      |       |      |      |      |      |      |      |      |     |
|-----|------|------|-------|------|------|-------|------|-------|------|------|------|------|------|-------|------|------|------|------|------|-------|------|------|------|------|-------|------|------|------|------|------|------|------|-----|
| hex | 00C6 | 00D0 | 00D8  | 00DE | 00DF | 00E6  | 00F0 | 00F8  | 00FE | 0110 | 0111 | 0126 | 0127 | 0131  | 0141 | 0142 | 0152 | 0153 | 017F | 0181  | 0182 | 0186 | 0187 | 0189 | 018A  | 018B | 018E | 018F | 0190 | 0191 | 0192 | 0193 |     |
| reg | ÆĖĖ  | ÐÐÐ  | ÐÐÐ   | ÐÐÐ  | ÐÐÐ  | ßßß   | æææ  | ððð   | øøø  | þþþ  | ÐÐÐ  | ddd  | HHH  | hhh   | ii   | LLL  | lll  | Œœœ  | œœœ  | ſſſ   | B̢B̢ | B̢B̢ | ᶑᶑᶑ  | CCC  | ÐÐÐ   | ÐÐÐ  | ÐÐÐ  | ÐÐÐ  | ÐÐÐ  | ÐÐÐ  | ÐÐÐ  | ÐÐÐ  | ÐÐÐ |
| ax  | 480  | 341  | 351   | 281  |      | 350   |      |       |      | 278  |      |      |      | 132   |      |      | 460  | 405  |      | 327   |      |      |      |      | 306   |      |      |      |      |      |      |      |     |
| ay  | 850  | 850  | 850   | 850  |      | 645   |      |       |      | 646  |      |      |      | 645   |      |      | 850  | 645  |      | 805   |      |      |      |      | 805   |      |      |      |      |      |      |      |     |
| bx  | 549  |      | 350   |      |      | 327   |      | 243   |      |      |      |      |      | 139   |      |      |      |      |      | 328   |      |      |      |      | 369   |      |      |      |      |      |      |      |     |
| by  | -107 |      | -110  |      |      | -109  |      | -110  |      |      |      |      |      | -108  |      |      |      |      |      | -108  |      |      |      |      | -107  |      |      |      |      |      |      |      |     |
| □xm | 426  | 333  | 351   | 249  | 259  | 343   | 240  | 250   | 236  | 336  | 270  | 362  | 270  | 142   | 256  | 133  | 436  | 387  | 190  | 303   | 273  | 322  | 369  | 333  | 368   | 292  | 280  | 326  | 256  | 183  | 201  | 371  |     |
| axδ | 53   | 8    | 0     | 31   |      | 6     |      |       |      | 41   |      |      |      | -10   |      |      | 24   | 17   |      | 24    |      |      |      |      | -62   |      |      |      |      |      |      |      |     |
| ax% | 0.06 | 0.01 | -0.00 | 0.06 |      | 0.01  |      |       |      | 0.09 |      |      |      | -0.04 |      |      | 0.03 | 0.02 |      | 0.04  |      |      |      |      | -0.09 |      |      |      |      |      |      |      |     |
| bxδ | 122  |      | -1    |      |      | -16   |      |       |      | -7   |      |      |      | -3    |      |      |      |      |      | 25    |      |      |      |      | 0     |      |      |      |      |      |      |      |     |
| bx% | 0.14 |      | -0.00 |      |      | -0.03 |      | -0.02 |      |      |      |      |      | -0.01 |      |      |      |      |      | 0.04  |      |      |      |      | 0.00  |      |      |      |      |      |      |      |     |
| it  | ÆĖĖ  | ÐÐÐ  | ÐÐÐ   | ÐÐÐ  | ÐÐÐ  | ßßß   | æææ  | ððð   | øøø  | þþþ  | ÐÐÐ  | ddd  | HHH  | hhh   | ii   | LLL  | lll  | Œœœ  | œœœ  | ſſſ   | B̢B̢ | B̢B̢ | ᶑᶑᶑ  | CCC  | ÐÐÐ   | ÐÐÐ  | ÐÐÐ  | ÐÐÐ  | ÐÐÐ  | ÐÐÐ  | ÐÐÐ  | ÐÐÐ  | ÐÐÐ |
| ax  | 615  |      |       |      |      | 480   |      |       |      | 278  |      |      |      | 267   |      |      | 575  | 480  |      | 327   |      |      |      |      | 306   |      |      |      |      |      |      |      |     |
| ay  | 850  |      |       |      |      | 645   |      |       |      | 646  |      |      |      | 645   |      |      | 850  | 645  |      | 805   |      |      |      |      | 805   |      |      |      |      |      |      |      |     |
| bx  | 549  |      | 310   |      |      | 327   |      | 186   |      |      |      |      |      | 139   |      |      |      |      |      | 328   |      |      |      |      | 369   |      |      |      |      |      |      |      |     |
| by  | -107 |      | -113  |      |      | -109  |      | -107  |      |      |      |      |      | -108  |      |      |      |      |      | -108  |      |      |      |      | -107  |      |      |      |      |      |      |      |     |
| □xm | 433  | 355  | 411   | 274  | 244  | 394   | 298  | 283   | 247  | 355  | 333  | 398  | 308  | 197   | 268  | 199  | 481  | 399  | 217  | 347   | 314  | 386  | 483  | 375  | 433   | 358  | 377  | 411  | 307  | 228  | 268  | 462  |     |
| axδ | 181  |      |       |      |      | 85    |      |       |      | 30   |      |      |      | 70    |      |      | 94   | 81   |      | -20   |      |      |      |      | -127  |      |      |      |      |      |      |      |     |
| ax% | 0.21 |      |       |      |      | 0.13  |      |       |      | 0.06 |      |      |      | 0.39  |      |      | 0.12 | 0.13 |      | -0.04 |      |      |      |      | -0.19 |      |      |      |      |      |      |      |     |
| bxδ | 115  |      | -101  |      |      | -67   |      | -97   |      |      |      |      |      | -58   |      |      |      |      |      | -19   |      |      |      |      | -64   |      |      |      |      |      |      |      |     |
| bx% | 0.13 |      | -0.15 |      |      | -0.10 |      | -0.22 |      |      |      |      |      | -0.32 |      |      |      |      |      | -0.03 |      |      |      |      | -0.09 |      |      |      |      |      |      |      |     |

Table 4: Font metrics for Latin and IPA 2/4

| hex | 0194  | 0196 | 0197 | 0198 | 019C   | 019D | 019F  | 01B7  | 0237  | 0250  | 0251  | 0252  | 0253  | 0254  | 0255  | 0256  | 0257  | 0258  | 0259  | 025A  | 025B  | 025C  | 025D  | 025E  | 025F  | 0260  | 0261  | 0262  | 0263  | 0264  | 0265  | 0266  |
|-----|-------|------|------|------|--------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| reg | ŸŸŸ   | lll  | lll  | KKK  | llllll | NNN  | 000   | 333   | jjj   | eee   | aaa   | ooo   | bbb   | ddd   | ccc   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   |
| ax  | 331   |      |      |      | 431    |      |       |       | 140   | 228   | 228   | 264   | 480   | 216   | 209   | 218   | 218   | 204   | 195   | 229   | 209   | 194   | 209   | 239   | 169   | 209   | 239   | 265   | 259   | 248   | 268   | 299   |
| ay  | 805   |      |      |      | 801    |      |       |       | 645   | 645   | 645   | 645   | 630   | 645   | 645   | 645   | 645   | 645   | 645   | 645   | 645   | 645   | 645   | 645   | 645   | 645   | 645   | 645   | 645   | 645   | 565   |       |
| bx  |       |      |      |      | 450    |      | 350   | 272   |       | 229   | 212   | 259   | 212   | 177   | 212   | 212   | 212   | 212   | 186   | 212   | 212   | 184   | 212   | 256   | 196   | 236   | 226   | 248   | 251   | 216   | 219   | 240   |
| by  |       |      |      |      | -108   |      | -110  | -319  |       | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -110  | -110  | -320  | -110  | -277  | -114  | -129  | -114  |
| □xm | 308   | 151  | 148  | 339  | 433    | 324  | 351   | 303   | 74    | 216   | 272   | 247   | 265   | 215   | 222   | 314   | 322   | 220   | 232   | 306   | 201   | 195   | 234   | 241   | 124   | 320   | 228   | 259   | 258   | 221   | 271   | 262   |
| axδ | 22    |      |      |      | -2     |      |       |       | 66    | 11    | -44   | 17    | 215   | 0     | -13   | -96   | -104  | -16   | -37   | -77   | 7     | -1    | -25   | -2    | 44    | -111  | 11    | 6     | 1     | 27    | -3    | 36    |
| ax% | 0.04  |      |      |      | -0.00  |      |       |       | 0.27  | 0.03  | -0.09 | 0.04  | 0.56  | 0.00  | -0.04 | -0.17 | -0.18 | -0.04 | -0.10 | -0.14 | 0.02  | -0.00 | -0.06 | -0.01 | 0.13  | -0.20 | 0.03  | 0.01  | 0.00  | 0.06  | -0.01 | 0.07  |
| bxδ |       |      |      |      | 16     |      | -1    | -31   |       | 12    | -60   | 12    | -53   | -38   | -10   | -102  | -110  | -8    | -46   | -94   | 10    | -11   | -22   | 14    | 71    | -84   | -2    | -11   | -7    | -5    | -52   | -22   |
| bx% |       |      |      |      | 0.02   |      | -0.00 | -0.06 |       | 0.03  | -0.13 | 0.03  | -0.14 | -0.11 | -0.03 | -0.19 | -0.19 | -0.02 | -0.12 | -0.17 | 0.03  | -0.04 | -0.06 | 0.03  | 0.21  | -0.15 | -0.01 | -0.02 | -0.01 | -0.01 | -0.10 | -0.05 |
| it  | ŸŸŸ   | lll  | lll  | KKK  | llllll | NNN  | 000   | 333   | jjj   | eee   | aaa   | ooo   | bbb   | ddd   | ccc   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   |
| ax  | 331   |      |      |      | 431    |      |       |       | 140   |       |       |       |       | 322   |       | 218   | 218   | 204   |       |       | 322   |       |       |       |       |       | 265   |       |       | 258   | 299   |       |
| ay  | 805   |      |      |      | 801    |      |       |       | 645   |       |       |       |       | 654   |       | 645   | 645   | 645   |       |       | 654   |       |       |       |       |       | 645   |       |       | 645   | 565   |       |
| bx  |       |      |      |      | 450    |      |       |       |       | 229   | 212   | 243   | 212   | 172   | 212   | 212   | 212   | 212   | 212   | 212   | 212   | 184   | 212   | 256   |       |       | 248   |       | 177   | 219   | 240   |       |
| by  |       |      |      |      | -108   |      |       |       |       | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  |       |       | -114  |       | -114  | -129  | -114  |       |
| □xm | 392   | 202  | 203  | 375  | 536    | 368  | 405   | 341   | 96    | 299   | 297   | 303   | 286   | 236   | 253   | 325   | 396   | 260   | 268   | 372   | 252   | 228   | 315   | 292   | 157   | 402   | 279   | 291   | 346   | 297   | 313   | 309   |
| axδ | -61   |      |      |      | -105   |      |       |       | 43    |       |       |       |       | 86    |       | -107  | -178  | -56   |       |       | 69    |       |       |       |       |       | -26   |       |       | -55   | -10   |       |
| ax% | -0.10 |      |      |      | -0.13  |      |       |       | 0.12  |       |       |       |       | 0.25  |       | -0.21 | -0.28 | -0.16 |       |       | 0.19  |       |       |       |       |       | -0.06 |       |       | -0.13 | -0.02 |       |
| bxδ |       |      |      |      | -86    |      |       |       |       | -70   | -85   | -60   | -74   | -64   | -41   | -113  | -184  | -48   | -56   | -160  | -40   | -44   | -103  | -36   |       |       | -43   |       | -120  | -94   | -69   |       |
| bx% |       |      |      |      | -0.10  |      |       |       |       | -0.17 | -0.19 | -0.14 | -0.17 | -0.19 | -0.11 | -0.23 | -0.29 | -0.14 | -0.16 | -0.27 | -0.11 | -0.12 | -0.19 | -0.08 |       |       | -0.10 |       | -0.34 | -0.22 | -0.16 |       |
| sb  | ŸŸŸ   | lll  | lll  | KKK  | llllll | NNN  | 000   | 333   | jjj   | eee   | aaa   | ooo   | bbb   | ddd   | ccc   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   |
| ax  |       |      |      |      |        |      |       |       | 163   |       |       |       |       | 199   |       |       |       |       |       |       | 229   |       |       |       |       |       |       |       |       |       |       |       |
| ay  |       |      |      |      |        |      |       |       | 645   |       |       |       |       | 645   |       |       |       |       |       |       | 645   |       |       |       |       |       |       |       |       |       |       |       |
| bx  |       |      |      |      |        |      |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| by  |       |      |      |      |        |      |       |       | 272   |       | 228   |       |       | 184   |       |       |       |       | 212   |       |       |       |       |       |       |       |       |       |       |       |       |       |
| □xm | 293   | 169  | 158  | 337  | 444    | 336  | 365   | 315   | 78    | 207   | 276   | 251   | 268   | 199   | 232   | 322   | 323   | 230   | 228   | 319   | 212   | 215   | 276   | 249   | 144   | 336   | 235   | 265   | 272   | 202   | 277   | 274   |
| axδ |       |      |      |      |        |      |       |       | 85    |       |       |       |       | 0     |       |       |       |       |       |       | 16    |       |       |       |       |       |       |       |       |       |       |       |
| ax% |       |      |      |      |        |      |       |       | 0.30  |       |       |       |       | 0.00  |       |       |       |       |       |       | 0.05  |       |       |       |       |       |       |       |       |       |       |       |
| bxδ |       |      |      |      |        |      |       |       | -43   |       | 21    |       |       | -15   |       |       |       |       | -16   |       |       |       |       |       |       |       |       |       |       |       |       |       |
| bx% |       |      |      |      |        |      |       |       | -0.08 |       | 0.05  |       |       | -0.04 |       |       |       |       | -0.04 |       |       |       |       |       |       |       |       |       |       |       |       |       |
| si  | ŸŸŸ   | lll  | lll  | KKK  | llllll | NNN  | 000   | 333   | jjj   | eee   | aaa   | ooo   | bbb   | ddd   | ccc   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   | ddd   |
| ax  | 331   |      |      |      | 431    |      |       |       | 140   |       |       |       |       | 160   |       | 218   | 218   | 204   |       |       | 209   | 194   |       |       |       |       | 265   |       |       | 258   | 299   |       |
| ay  | 805   |      |      |      | 801    |      |       |       | 645   |       |       |       |       | 645   |       | 645   | 645   | 645   |       |       | 645   | 645   |       |       |       |       | 645   |       |       | 645   | 565   |       |
| bx  |       |      |      |      | 450    |      |       |       | 236   |       | 229   | 212   | 259   | 212   | 177   | 212   | 212   | 212   |       |       | 212   | 212   | 184   | 212   | 256   |       | 248   |       | 177   | 219   | 240   |       |
| by  |       |      |      |      | -108   |      |       |       | -319  |       | -114  | -114  | -114  | -114  | -114  | -114  | -114  | -114  |       |       | -114  | -114  | -114  | -114  | -114  |       | -110  |       | -114  | -129  | -114  |       |
| □xm | 405   | 226  | 227  | 388  | 547    | 380  | 436   | 341   | 135   | 290   | 310   | 317   | 297   | 237   | 292   | 339   | 416   | 266   | 278   | 383   | 265   | 250   | 334   | 308   | 164   | 415   | 288   | 311   | 369   | 298   | 369   | 277   |
| axδ | -74   |      |      |      | -116   |      |       |       | 5     |       | -82   | -53   |       | -77   |       | -121  | -198  | -62   |       |       | -56   | -56   |       |       |       |       | -46   |       | -111  | 21    |       |       |
| ax% | -0.11 |      |      |      | -0.13  |      |       |       | 0.01  |       | -0.17 | -0.11 |       | -0.19 |       | -0.22 | -0.28 | -0.14 |       |       | -0.14 | -0.14 |       |       |       |       | -0.10 |       | -0.21 | 0.04  |       |       |
| bxδ |       |      |      |      | -97    |      |       |       | -105  |       | -61   | -98   | -58   | -85   | -60   | -80   | -127  | -204  | -54   |       | -171  | -53   | -66   | -122  | -52   |       | -63   |       | -121  | -150  | -37   |       |
| bx% |       |      |      |      | -0.11  |      |       |       | -0.16 |       | -0.14 | -0.20 | -0.12 | -0.18 | -0.15 | -0.18 | -0.24 | -0.29 | -0.12 |       | -0.27 | -0.13 | -0.17 | -0.22 | -0.11 |       | -0.13 |       | -0.31 | -0.29 | -0.07 |       |

Table 5: Font metrics for Latin and IPA 3/4

| hex | 0267  | 0268  | 0269  | 026A  | 026B  | 026C  | 026D | 026E  | 026F  | 0270  | 0271  | 0272 | 0273  | 0274  | 0275  | 0277  | 0278  | 0279  | 027A  | 027B  | 027C  | 027D | 027E | 027F | 0280  | 0281  | 0282  | 0283 | 0284 | 0285 | 0286 | 0287 |
|-----|-------|-------|-------|-------|-------|-------|------|-------|-------|-------|-------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|-------|-------|-------|------|------|------|------|------|
| reg | ḡḡḡ   | ḡḡḡ   | ḡḡ    | ḡḡ    | ḡḡ    | ḡḡ    | ḡḡ   | ḡḡḡ   | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ |      |
| ax  | 299   | 155   | 123   | 133   | 209   | 209   | 209  | 209   | 407   | 209   | 383   | 264  | 263   | 209   | 209   | 209   | 209   | 209   | 209   | 209   | 222   | 209  | 209  | 209  | 209   | 209   | 209   | 209  | 209  | 209  | 209  |      |
| ay  | 565   | 795   | 645   | 645   | 645   | 645   | 645  | 645   | 645   | 645   | 645   | 646  | 645   | 645   | 645   | 645   | 645   | 645   | 645   | 645   | 703   | 645  | 645  | 645  | 645   | 645   | 645   | 645  | 645  | 645  | 645  |      |
| bx  | 240   | 134   | 153   | 139   | 146   | 218   | 196  | 196   | 410   | 294   | 383   | 265  | 262   | 247   | 196   | 341   | 196   | 190   | 203   | 196   | 127   | 196  | 196  | 196  | 196   | 196   | 196   | 196  | 196  | 196  | 196  |      |
| by  | -114  | -114  | -114  | -114  | -114  | -120  | -110 | -110  | -114  | -114  | -105  | -111 | -112  | -114  | -110  | -114  | -110  | -114  | -114  | -110  | -305  | -110 | -110 | -110 | -110  | -110  | -110  | -110 | -110 | -110 | -110 |      |
| □xm | 227   | 149   | 170   | 136   | 144   | 182   | 182  | 250   | 390   | 379   | 365   | 234  | 295   | 278   | 252   | 357   | 285   | 177   | 179   | 225   | 186   | 186  | 174  | 170  | 238   | 238   | 199   | 140  | 140  | 170  | 149  | 150  |
| axδ | 71    | 5     | -47   | -3    | 65    | 27    | 27   | -41   | 16    | -170  | 17    | 29   | -32   | -69   | -43   | -148  | -76   | 31    | 29    |       | 35    | 22   | 34   | 38   | -29   | -29   | 10    | 68   | 68   | 38   | 59   | 59   |
| ax% | 0.17  | 0.02  | -0.23 | -0.01 | 0.23  | 0.08  | 0.09 | -0.09 | 0.02  | -0.23 | 0.03  | 0.05 | -0.06 | -0.14 | -0.10 | -0.23 | -0.15 | 0.09  | 0.09  |       | 0.11  | 0.07 | 0.11 | 0.12 | -0.07 | -0.07 | 0.03  | 0.17 | 0.17 | 0.09 | 0.15 | 0.21 |
| bxδ | 12    | -15   | -17   | 3     | 2     | 36    | 14   | -54   | 19    | -85   | 17    | 30   | -33   | -31   | -56   | -16   | -89   | 12    | 23    | -29   | -59   | 9    | 21   | 25   | -42   | -42   | -3    | 55   | 55   | 25   | 46   | 46   |
| bx% | 0.03  | -0.06 | -0.08 | 0.01  | 0.01  | 0.11  | 0.04 | -0.12 | 0.03  | -0.12 | 0.03  | 0.06 | -0.06 | -0.06 | -0.13 | -0.02 | -0.18 | 0.04  | 0.07  | -0.07 | -0.19 | 0.03 | 0.07 | 0.08 | -0.10 | -0.10 | -0.01 | 0.14 | 0.14 | 0.06 | 0.12 | 0.17 |
| it  | ḡḡḡ   | ḡḡḡ   | ḡḡ    | ḡḡ    | ḡḡ    | ḡḡ    | ḡḡ   | ḡḡḡ   | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ |      |
| ax  |       |       |       |       |       |       |      |       | 407   |       | 383   | 264  | 263   |       |       |       |       |       |       |       | 222   |      |      |      |       |       |       |      |      |      |      |      |
| ay  |       |       |       |       |       |       |      |       | 645   |       | 645   | 646  | 645   |       |       |       |       |       |       |       | 703   |      |      |      |       |       |       |      |      |      |      |      |
| bx  |       | 134   | 153   | 139   | 146   | 218   |      |       | 410   | 294   | 383   | 265  | 262   | 247   |       | 341   |       | 190   | 203   |       | 127   |      |      |      |       |       |       |      |      |      |      |      |
| by  |       | -114  | -114  | -114  | -114  | -120  |      |       | -114  | -114  | -105  | -111 | -112  | -114  |       | -114  |       | -114  | -114  |       | -305  |      |      |      |       |       |       |      |      |      |      |      |
| □xm | 297   | 204   | 178   | 186   | 210   | 240   | 193  | 284   | 472   | 473   | 388   | 210  | 295   | 328   | 289   | 398   | 342   | 98    | 268   | 233   | 217   | 265  | 218  | 215  | 245   | 260   | 217   | 197  | 197  | 198  | 206  | 170  |
| axδ |       |       | -55   | -58   |       |       |      |       | -65   |       | -5    | 53   | -32   |       |       |       |       |       |       |       | 5     |      |      |      |       |       |       |      |      |      |      |      |
| ax% |       |       | -0.31 | -0.18 |       |       |      |       | -0.09 |       | -0.01 | 0.09 | -0.08 |       |       |       |       |       |       |       | 0.01  |      |      |      |       |       |       |      |      |      |      |      |
| bxδ |       | -70   | -25   | -47   | -64   | -22   |      |       | -62   | -179  | -5    | 54   | -33   | -81   |       | -57   |       | 92    | -65   |       | -90   |      |      |      |       |       |       |      |      |      |      |      |
| bx% |       | -0.28 | -0.14 | -0.15 | -0.22 | -0.07 |      |       | -0.09 | -0.25 | -0.01 | 0.09 | -0.08 | -0.14 |       | -0.09 |       | 0.27  | -0.13 |       | -0.19 |      |      |      |       |       |       |      |      |      |      |      |
| sb  | ḡḡḡ   | ḡḡḡ   | ḡḡ    | ḡḡ    | ḡḡ    | ḡḡ    | ḡḡ   | ḡḡḡ   | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ |      |
| ax  |       |       |       |       |       |       |      |       | 377   |       |       |      |       |       |       |       |       |       |       |       |       |      |      |      |       |       |       |      |      |      |      |      |
| ay  |       |       |       |       |       |       |      |       | 645   |       |       |      |       |       |       |       |       |       |       |       |       |      |      |      |       |       |       |      |      |      |      |      |
| bx  |       | 134   |       | 156   | 146   | 218   |      |       | 380   |       |       |      |       |       | 229   |       |       | 190   |       |       |       |      |      |      |       |       |       |      |      |      |      |      |
| by  |       | -114  |       | -114  | -114  | -120  |      |       | -114  |       |       |      |       |       | -114  |       |       | -114  |       |       |       |      |      |      |       |       |       |      |      |      |      |      |
| □xm | 243   | 152   | 164   | 152   | 156   | 190   | 193  | 273   | 354   | 388   | 363   | 245  | 306   | 290   | 259   | 369   | 297   | 144   | 205   | 237   | 205   | 198  | 186  | 182  | 235   | 220   | 201   | 152  | 152  | 152  | 161  | 157  |
| axδ |       |       |       | -6    |       |       |      |       | 23    |       |       |      |       |       |       |       |       |       |       |       |       |      |      |      |       |       |       |      |      |      |      |      |
| ax% |       |       |       | -0.02 |       |       |      |       | 0.03  |       |       |      |       |       |       |       |       |       |       |       |       |      |      |      |       |       |       |      |      |      |      |      |
| bxδ |       | -18   |       | 4     | -10   | 27    |      |       | 26    |       |       |      |       |       | -30   |       |       | 45    |       |       |       |      |      |      |       |       |       |      |      |      |      |      |
| bx% |       | -0.07 |       | 0.02  | -0.04 | 0.08  |      |       | 0.03  |       |       |      |       |       | -0.07 |       |       | 0.13  |       |       |       |      |      |      |       |       |       |      |      |      |      |      |
| si  | ḡḡḡ   | ḡḡḡ   | ḡḡ    | ḡḡ    | ḡḡ    | ḡḡ    | ḡḡ   | ḡḡḡ   | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ  | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ | ḡḡḡḡ |      |
| ax  | 299   | 137   | 123   | 128   |       |       |      |       | 407   |       | 383   | 264  | 263   |       |       |       |       |       |       |       | 222   |      |      |      |       |       |       |      |      |      |      |      |
| ay  | 565   | 795   | 645   | 645   |       |       |      |       | 645   |       | 645   | 646  | 645   |       |       |       |       |       |       |       | 703   |      |      |      |       |       |       |      |      |      |      |      |
| bx  | 240   | 136   | 153   | 139   | 146   | 218   |      |       | 410   | 294   | 383   | 265  | 262   | 247   |       | 341   |       | 190   | 203   |       | 127   |      |      |      |       |       |       |      |      |      |      |      |
| by  | -114  | -110  | -114  | -114  | -114  | -120  |      |       | -114  | -114  | -105  | -111 | -112  | -114  |       | -114  |       | -114  | -114  |       | -305  |      |      |      |       |       |       |      |      |      |      |      |
| □xm | 279   | 208   | 215   | 198   | 227   | 288   | 235  | 298   | 467   | 484   | 401   | 227  | 321   | 340   | 321   | 412   | 347   | 225   | 280   | 246   | 230   | 272  | 230  | 228  | 257   | 274   | 244   | 209  | 209  | 208  | 215  | 162  |
| axδ | 19    | -71   | -92   | -70   |       |       |      |       | -60   |       | -18   | 37   | -58   |       |       |       |       |       |       |       | -8    |      |      |      |       |       |       |      |      |      |      |      |
| ax% | 0.04  | -0.27 | -0.39 | -0.20 |       |       |      |       | -0.08 |       | -0.02 | 0.06 | -0.09 |       |       |       |       |       |       |       | -0.02 |      |      |      |       |       |       |      |      |      |      |      |
| bxδ | -39   | -72   | -62   | -59   | -81   | -70   |      |       | -57   | -190  | -18   | 38   | -59   | -93   |       | -71   |       | -35   | -77   |       | -103  |      |      |      |       |       |       |      |      |      |      |      |
| bx% | -0.08 | -0.28 | -0.26 | -0.17 | -0.23 | -0.20 |      |       | -0.08 | -0.25 | -0.02 | 0.06 | -0.10 | -0.15 |       | -0.10 |       | -0.08 | -0.15 |       | -0.20 |      |      |      |       |       |       |      |      |      |      |      |

Table 6: Font metrics for Latin and IPA 4/4

| hex | 0288       | 0289       | 028A       | 028B       | 028C       | 028D       | 028E       | 028F       | 0290       | 0291       | 0292       | 0293       | 0294       | 0295       | 0296       | 0297       | 0299       | 029B       | 029C       | 029D       | 029F       |
|-----|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| reg | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> |
| ax  | 209        | 209        | 209        | 259        | 209        | 209        | 209        | 209        | 209        | 209        | 218        | 209        | 209        | 209        | 209        | 209        | 209        | 209        | 209        | 209        | 209        |
| ay  | 645        | 645        | 645        | 645        | 645        | 645        | 645        | 645        | 645        | 645        | 646        | 645        | 645        | 645        | 645        | 645        | 645        | 645        | 645        | 645        | 645        |
| bx  | 196        | 196        | 258        | 251        | 196        | 196        | 196        | 239        | 196        | 196        | 185        | 196        | 202        | 230        | 196        | 196        | 196        | 196        | 305        | 196        | 196        |
| by  | -110       | -110       | -114       | -114       | -110       | -110       | -110       | -107       | -110       | -110       | -319       | -110       | -107       | -107       | -110       | -110       | -110       | -110       | -110       | -110       | -110       |
| □xm | 157        | 271        | 253        | 235        | 244        | 367        | 248        | 236        | 262        | 217        | 216        | 220        | 211        | 221        | 208        | 221        | 232        | 294        | 293        | 126        | 207        |
| axδ | 51         | -62        | -44        | 24         | -35        | -158       | -39        | -27        | -53        | -8         | 1          | -11        | -2         | -12        | 1          | -12        | -23        | -85        | -84        | 83         | 2          |
| ax% | 0.19       | -0.12      | -0.09      | 0.06       | -0.07      | -0.22      | -0.08      | -0.06      | -0.12      | -0.02      | 0.00       | -0.03      | -0.01      | -0.03      | 0.00       | -0.03      | -0.06      | -0.16      | -0.16      | 0.29       | 0.01       |
| bxδ | 38         | -75        | 4          | 16         | -48        | -171       | -52        | 3          | -66        | -21        | -31        | -24        | -9         | 9          | -12        | -25        | -36        | -98        | 12         | 70         | -11        |
| bx% | 0.14       | -0.15      | 0.01       | 0.04       | -0.10      | -0.23      | -0.10      | 0.01       | -0.15      | -0.06      | -0.09      | -0.07      | -0.02      | 0.02       | -0.03      | -0.07      | -0.09      | -0.19      | 0.02       | 0.24       | -0.03      |
| it  | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> |
| ax  |            |            |            | 259        |            |            |            |            |            |            | 328        |            |            |            |            |            |            |            |            |            |            |
| ay  |            |            |            | 645        |            |            |            |            |            |            | 646        |            |            |            |            |            |            |            |            |            |            |
| bx  |            |            | 258        | 251        |            |            |            | 239        |            |            | 130        |            | 202        | 230        |            |            |            |            | 273        |            |            |
| by  |            |            | -114       | -114       |            |            |            | -107       |            |            | -319       |            | -107       | -107       |            |            |            |            | -110       |            |            |
| □xm | 224        | 318        | 312        | 323        | 248        | 372        | 305        | 332        | 251        | 251        | 236        | 246        | 286        | 293        | 242        | 290        | 245        | 361        | 330        | 145        | 222        |
| axδ |            |            |            | -64        |            |            |            |            |            |            | 92         |            |            |            |            |            |            |            |            |            |            |
| ax% |            |            |            | -0.15      |            |            |            |            |            |            | 0.22       |            |            |            |            |            |            |            |            |            |            |
| bxδ |            |            | -54        | -72        |            |            |            | -93        |            |            | -106       |            | -84        | -63        |            |            |            |            | -57        |            |            |
| bx% |            |            | -0.11      | -0.17      |            |            |            | -0.21      |            |            | -0.25      |            | -0.21      | -0.16      |            |            |            |            | -0.10      |            |            |
| sb  | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> |
| ax  |            |            | 266        |            |            |            |            |            |            |            | 234        |            |            |            |            |            |            |            |            |            |            |
| ay  |            |            | 645        |            |            |            |            |            |            |            | 645        |            |            |            |            |            |            |            |            |            |            |
| bx  |            |            |            |            |            |            |            |            |            |            | 193        |            |            |            |            |            |            |            | 351        |            |            |
| by  |            |            |            |            |            |            |            |            |            |            | -319       |            |            |            |            |            |            |            | -110       |            |            |
| □xm | 182        | 274        | 267        | 255        | 231        | 355        | 243        | 244        | 254        | 229        | 227        | 236        | 206        | 206        | 206        | 273        | 245        | 290        | 316        | 132        | 219        |
| axδ |            |            | -1         |            |            |            |            |            |            |            | 7          |            |            |            |            |            |            |            |            |            |            |
| ax% |            |            | -0.00      |            |            |            |            |            |            |            | 0.02       |            |            |            |            |            |            |            |            |            |            |
| bxδ |            |            |            |            |            |            |            |            |            |            | -34        |            |            |            |            |            |            |            | 35         |            |            |
| bx% |            |            |            |            |            |            |            |            |            |            | -0.08      |            |            |            |            |            |            |            | 0.06       |            |            |
| si  | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> | <i>ttt</i> |
| ax  |            |            |            | 259        |            |            |            |            |            |            | 218        |            |            |            |            |            |            |            |            |            |            |
| ay  |            |            |            | 645        |            |            |            |            |            |            | 646        |            |            |            |            |            |            |            |            |            |            |
| bx  |            |            | 258        | 251        |            |            |            | 239        |            |            | 141        |            | 202        | 230        |            |            |            |            | 327        |            |            |
| by  |            |            | -114       | -114       |            |            |            | -107       |            |            | -319       |            | -107       | -107       |            |            |            |            | -110       |            |            |
| □xm | 237        | 349        | 325        | 336        | 241        | 367        | 303        | 341        | 281        | 264        | 237        | 259        | 297        | 306        | 255        | 303        | 267        | 390        | 404        | 159        | 235        |
| axδ |            |            |            | -77        |            |            |            |            |            |            | -19        |            |            |            |            |            |            |            |            |            |            |
| ax% |            |            |            | -0.17      |            |            |            |            |            |            | -0.04      |            |            |            |            |            |            |            |            |            |            |
| bxδ |            |            | -67        | -85        |            |            |            | -102       |            |            | -96        |            | -95        | -76        |            |            |            |            | -77        |            |            |
| bx% |            |            | -0.12      | -0.19      |            |            |            | -0.22      |            |            | -0.19      |            | -0.21      | -0.18      |            |            |            |            | -0.11      |            |            |



Table 7: Font metrics for above marks

| hex | 0300  | 0301  | 0302  | 0303  | 0304  | 0306  | 0307  | 0308  | 0309  | 030A  | 030B  | 030C  | 030D  | 030F  | 0310  | 0311  | 0312  | 033D  | 033E  | 033F | 030E  | 0346  | 034A | 034B | 034C | 0350  | 0351  | 0352 | 0357  | 035B |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|-------|-------|------|------|------|-------|-------|------|-------|------|
| reg | `     | ´     | ^     | ~     | _     | ˘     | ˙     | ˚     | ˛     | ˜     | ˝     | ˘     | ˙     | ˚     | ˛     | ˜     | ˝     | ˘     | ˙     | ˚    | ˛     | ˜     | ˝    | ˘    | ˙    | ˚     | ˛     | ˜    | ˝     |      |
| ax  | -142  | -191  | -139  | -149  | -209  | -161  | -180  | -210  | -160  | -184  | -219  | -205  | -155  | -140  | -251  | -205  | -110  | -265  | -265  | -235 | -245  | -251  | -215 | -205 | -235 | -201  | -171  | -251 | -171  | -164 |
| ay  | 683   | 682   | 697   | 697   | 644   | 681   | 667   | 645   | 765   | 708   | 714   | 676   | 688   | 758   | 675   | 645   | 605   | 605   | 605   | 611  | 719   | 621   | 645  | 645  | 645  | 645   | 645   | 645  | 645   | 645  |
| ◻xm | -159  | -160  | -140  | -149  | -212  | -161  | -180  | -210  | -161  | -182  | -170  | -204  | -154  | -168  | -251  | -208  | -102  | -261  | -247  | -240 | -245  | -255  | -217 | -217 | -239 | -183  | -156  | -251 | -141  | -175 |
| axδ | 17    | -31   | 1     | 0     | 2     | 0     | 0     | 0     | 1     | -2    | -49   | -1    | -1    | 27    | 0     | 3     | -8    | -4    | -18   | 5    | 0     | 4     | 2    | 12   | 3    | -18   | -15   | 0    | -30   | 10   |
| ax% | 0.11  | -0.20 | 0.00  | 0.00  | 0.01  | 0.00  | 0.00  | -0.00 | 0.01  | -0.01 | -0.19 | -0.01 | -0.02 | 0.11  | 0.00  | 0.01  | -0.08 | -0.02 | -0.21 | 0.01 | -0.00 | 0.02  | 0.01 | 0.04 | 0.01 | -0.08 | -0.12 | 0.00 | -0.23 | 0.05 |
| it  | `     | ´     | ^     | ~     | _     | ˘     | ˙     | ˚     | ˛     | ˜     | ˝     | ˘     | ˙     | ˚     | ˛     | ˜     | ˝     | ˘     | ˙     | ˚    | ˛     | ˜     | ˝    | ˘    | ˙    | ˚     | ˛     | ˜    | ˝     |      |
| ax  | -3    | -52   | 3     | -17   | -78   | -22   | -40   | -79   | -4    | -40   | -74   | -67   | -15   | 14    | -114  | -74   | 13    | -142  | -142  | -111 | -99   | -125  | -84  | -74  | -104 | -70   | -40   | -120 | -40   | -33  |
| ay  | 683   | 682   | 697   | 697   | 645   | 681   | 689   | 645   | 765   | 708   | 714   | 676   | 688   | 758   | 675   | 645   | 605   | 605   | 605   | 611  | 719   | 621   | 645  | 645  | 645  | 645   | 645   | 645  | 645   | 645  |
| ◻xm | -33   | -34   | -24   | -23   | -98   | -26   | -55   | -94   | -14   | -49   | -41   | -69   | -26   | -35   | -116  | -104  | 16    | -151  | -134  | -129 | -117  | -146  | -99  | -97  | -119 | -71   | -31   | -131 | -25   | -52  |
| axδ | 29    | -18   | 27    | 5     | 19    | 4     | 15    | 15    | 9     | 8     | -33   | 1     | 11    | 48    | 2     | 30    | -3    | 8     | -8    | 18   | 17    | 20    | 14   | 22   | 15   | 0     | -9    | 11   | -15   | 18   |
| ax% | 0.22  | -0.10 | 0.11  | 0.02  | 0.07  | 0.01  | 0.16  | 0.06  | 0.06  | 0.05  | -0.12 | 0.01  | 0.13  | 0.20  | 0.01  | 0.11  | -0.03 | 0.04  | -0.08 | 0.04 | 0.09  | 0.08  | 0.05 | 0.08 | 0.05 | 0.00  | -0.07 | 0.04 | -0.11 | 0.09 |
| sb  | `     | ´     | ^     | ~     | _     | ˘     | ˙     | ˚     | ˛     | ˜     | ˝     | ˘     | ˙     | ˚     | ˛     | ˜     | ˝     | ˘     | ˙     | ˚    | ˛     | ˜     | ˝    | ˘    | ˙    | ˚     | ˛     | ˜    | ˝     |      |
| ax  | -127  | -180  | -127  | -133  | -205  | -149  |       |       |       |       | -197  | -193  | -137  | -127  | -239  | -205  |       |       |       |      | -232  |       |      |      |      |       |       |      |       | -180 |
| ay  | 713   | 713   | 700   | 679   | 625   | 684   |       |       |       |       | 733   | 676   | 719   | 758   | 675   | 645   |       |       |       |      | 719   |       |      |      |      |       |       |      |       | 639  |
| ◻xm | -145  | -144  | -128  | -137  | -200  | -149  | -168  | -186  | -149  | -170  | -158  | -193  | -137  | -153  | -235  | -220  | -102  |       |       |      |       | -232  |      |      |      |       | 105   |      | 111   | -184 |
| axδ | 17    | -36   | 0     | 3     | -5    | 0     |       |       |       |       | -39   | 0     | 0     | 25    | -4    | 15    |       |       |       |      | 0     |       |      |      |      |       |       |      |       | 4    |
| ax% | 0.10  | -0.21 | 0.00  | 0.01  | -0.02 | 0.00  |       |       |       |       | -0.15 | -0.00 | 0.00  | 0.10  | -0.01 | 0.05  |       |       |       |      | 0.00  |       |      |      |      |       |       |      |       | 0.02 |
| si  | `     | ´     | ^     | ~     | _     | ˘     | ˙     | ˚     | ˛     | ˜     | ˝     | ˘     | ˙     | ˚     | ˛     | ˜     | ˝     | ˘     | ˙     | ˚    | ˛     | ˜     | ˝    | ˘    | ˙    | ˚     | ˛     | ˜    | ˝     |      |
| ax  | -142  | -191  | -139  | -134  | -209  | -161  | -180  | -210  | -160  | -184  | -219  | -205  | -155  | -140  | -251  | -205  | -110  |       |       |      |       | -245  |      |      |      |       |       |      |       |      |
| ay  | 683   | 682   | 697   | 700   | 645   | 681   | 667   | 645   | 765   | 708   | 714   | 676   | 688   | 758   | 675   | 645   | 605   |       |       |      |       | 719   |      |      |      |       |       |      |       |      |
| ◻xm | -18   | -19   | -12   | -11   | -92   | -15   | -45   | -112  | -1    | -37   | -30   | -59   | -9    | -22   | -101  | -103  | 26    |       |       |      |       | -105  |      |      |      |       |       |      |       |      |
| axδ | -124  | -172  | -127  | -123  | -117  | -146  | -135  | -98   | -159  | -147  | -189  | -146  | -146  | -118  | -150  | -102  | -136  |       |       |      |       | -140  |      |      |      |       |       |      |       |      |
| ax% | -0.89 | -0.92 | -0.50 | -0.42 | -0.38 | -0.54 | -1.14 | -0.34 | -1.07 | -0.84 | -0.66 | -0.57 | -1.59 | -0.48 | -0.56 | -0.38 | -1.03 |       |       |      |       | -0.76 |      |      |      |       |       |      |       |      |

Table 8: Font metrics for below marks

| hex | 0316 | 0317 | 0318 | 0319  | 031C | 0320  | 0323 | 0324  | 0325  | 0326  | 0329  | 032A | 032B  | 032C  | 032D | 032E  | 032F | 031D  | 031E  | 031F | 0330  | 0331  | 0333  | 0339  | 033A | 033B | 033C | 0347  | 0348  | 0349  | 034D | 034E |
|-----|------|------|------|-------|------|-------|------|-------|-------|-------|-------|------|-------|-------|------|-------|------|-------|-------|------|-------|-------|-------|-------|------|------|------|-------|-------|-------|------|------|
| reg |      |      |      |       |      |       |      |       |       |       |       |      |       |       |      |       |      |       |       |      |       |       |       |       |      |      |      |       |       |       |      |      |
| bx  | -219 | -210 | -224 | -285  | -216 | -253  | -216 | -218  | -218  | -255  | -257  | -253 | -251  | -204  | -195 | -251  | -250 | -249  | -247  | -249 | -213  | -244  | -244  | -216  | -253 | -246 | -251 | -244  | -197  | -192  | -222 | -227 |
| by  | -133 | -132 | -148 | -147  | -146 | -148  | -113 | -116  | -116  | -82   | -122  | -113 | -108  | -116  | -133 | -110  | -116 | -143  | -106  | -145 | -70   | -110  | -110  | -146  | -113 | -113 | -108 | -110  | -110  | -110  | -110 | -110 |
| □xm | -248 | -225 | -249 | -249  | -262 | -252  | -216 | -217  | -217  | -245  | -256  | -255 | -252  | -204  | -195 | -252  | -251 | -249  | -249  | -252 | -205  | -240  | -240  | -178  | -255 | -252 | -252 | -240  | -195  | -189  | -229 | -229 |
| bxδ | 29   | 14   | 25   | -36   | 46   | -1    | 0    | -1    | -1    | -10   | -1    | 2    | 0     | 0     | 0    | 1     | 1    | 0     | 2     | 2    | -8    | -4    | -4    | -38   | 2    | 6    | 0    | -4    | -2    | -3    | 6    | 1    |
| bx% | 0.16 | 0.08 | 0.15 | -0.22 | 0.42 | -0.01 | 0.00 | -0.00 | -0.01 | -0.09 | -0.02 | 0.01 | 0.00  | -0.00 | 0.00 | 0.00  | 0.00 | 0.00  | 0.01  | 0.01 | -0.03 | -0.02 | -0.02 | -0.35 | 0.01 | 0.03 | 0.00 | -0.01 | -0.01 | -0.02 | 0.02 | 0.01 |
| it  |      |      |      |       |      |       |      |       |       |       |       |      |       |       |      |       |      |       |       |      |       |       |       |       |      |      |      |       |       |       |      |      |
| bx  | -246 | -237 | -254 | -315  | -246 | -283  | -239 | -242  | -242  | -272  | -282  | -276 | -273  | -228  | -222 | -275  | -274 | -278  | -269  | -278 | -227  | -266  | -266  | -246  | -276 | -269 | -273 | -266  | -219  | -214  | -244 | -249 |
| by  | -133 | -132 | -148 | -147  | -146 | -148  | -113 | -116  | -116  | -82   | -122  | -113 | -108  | -116  | -133 | -110  | -116 | -143  | -106  | -145 | -70   | -110  | -110  | -146  | -113 | -113 | -108 | -110  | -110  | -110  | -110 | -110 |
| □xm | -282 | -257 | -270 | -287  | -284 | -281  | -239 | -240  | -241  | -278  | -286  | -286 | -271  | -222  | -238 | -261  | -295 | -291  | -267  | -281 | -220  | -262  | -269  | -217  | -285 | -284 | -293 | -269  | -223  | -206  | -256 | -259 |
| bxδ | 35   | 20   | 16   | -28   | 37   | -2    | 0    | -2    | -1    | 5     | 4     | 10   | -2    | -6    | 16   | -14   | 20   | 13    | -2    | 3    | -7    | -4    | 3     | -29   | 8    | 14   | 19   | 3     | 4     | -8    | 11   | 10   |
| bx% | 0.21 | 0.11 | 0.09 | -0.16 | 0.29 | -0.01 | 0.00 | -0.01 | -0.01 | 0.04  | 0.05  | 0.04 | -0.01 | -0.03 | 0.06 | -0.05 | 0.08 | 0.07  | -0.01 | 0.02 | -0.02 | -0.02 | 0.01  | -0.23 | 0.03 | 0.07 | 0.06 | 0.01  | 0.02  | -0.05 | 0.03 | 0.05 |
| sb  |      |      |      |       |      |       |      |       |       |       |       |      |       |       |      |       |      |       |       |      |       |       |       |       |      |      |      |       |       |       |      |      |
| bx  | -219 | -210 | -224 | -285  | -216 | -253  | -216 | -218  | -218  | -255  | -257  | -253 | -251  | -208  | -195 | -239  | -250 | -249  | -247  | -249 | -213  | -244  |       |       |      |      |      |       |       |       |      |      |
| by  | -133 | -132 | -148 | -147  | -146 | -149  | -113 | -116  | -116  | -73   | -124  | -113 | -108  | -116  | -133 | -110  | -116 | -143  | -106  | -146 | -70   | -58   |       |       |      |      |      |       |       |       |      |      |
| □xm | -248 | -225 | -249 | -250  | -262 | -251  | -217 | -218  | -217  | -245  | -256  | -255 | -252  | -204  | -195 | -240  | -263 | -249  | -250  | -251 | -193  | -240  |       |       |      |      |      |       |       |       |      |      |
| bxδ | 28   | 14   | 24   | -35   | 46   | -2    | 0    | 0     | -1    | -10   | -1    | 2    | 0     | -4    | 0    | 1     | 13   | 0     | 2     | 2    | -20   | -4    |       |       |      |      |      |       |       |       |      |      |
| bx% | 0.16 | 0.08 | 0.15 | -0.22 | 0.42 | -0.01 | 0.00 | -0.00 | -0.01 | -0.09 | -0.02 | 0.01 | 0.00  | -0.02 | 0.00 | 0.00  | 0.05 | -0.00 | 0.01  | 0.01 | -0.07 | -0.01 |       |       |      |      |      |       |       |       |      |      |
| si  |      |      |      |       |      |       |      |       |       |       |       |      |       |       |      |       |      |       |       |      |       |       |       |       |      |      |      |       |       |       |      |      |
| bx  | -219 | -210 | -224 | -285  | -216 | -253  | -216 | -218  | -218  | -255  | -257  | -253 | -251  | -204  | -195 | -251  | -250 | -249  | -247  | -249 | -213  | -244  |       |       |      |      |      |       |       |       |      |      |
| by  | -133 | -132 | -148 | -147  | -146 | -148  | -113 | -116  | -116  | -82   | -122  | -113 | -108  | -116  | -133 | -110  | -116 | -143  | -106  | -145 | -70   | -110  |       |       |      |      |      |       |       |       |      |      |
| □xm | -270 | -245 | -258 | -288  | -271 | -269  | -227 | -228  | -228  | -261  | -275  | -274 | -259  | -211  | -226 | -249  | -294 | -281  | -252  | -269 | -196  | -239  |       |       |      |      |      |       |       |       |      |      |
| bxδ | 50   | 35   | 34   | 2     | 55   | 16    | 11   | 10    | 10    | 6     | 17    | 21   | 7     | 7     | 31   | -2    | 43   | 32    | 5     | 20   | -17   | -5    |       |       |      |      |      |       |       |       |      |      |
| bx% | 0.30 | 0.19 | 0.19 | 0.01  | 0.42 | 0.08  | 0.09 | 0.03  | 0.06  | 0.04  | 0.20  | 0.08 | 0.02  | 0.03  | 0.12 | -0.01 | 0.16 | 0.18  | 0.03  | 0.10 | -0.06 | -0.02 |       |       |      |      |      |       |       |       |      |      |

Consider a glyph that is not a full ascender. Find the nearest, or most suitable, vertical aspect – ascender  $a$ , capital  $c$ , or x-height  $x$ . Then  $ay = Y_{\max} + ay\delta[acx] - \epsilon$ , where  $Y_{\max}$  is the maximum  $Y$  coordinate of bounding box of the glyph, and  $\epsilon$  is the correction for the vertical aspect range, usually for an overshoot.

Consider a glyph that is not a full descender. Find the nearest, or most suitable, vertical aspect – baseline 0 or descender  $d$ . Then  $by = Y_{\min} + by\delta[0d] + \epsilon$ , where  $Y_{\min}$  is the minimum  $Y$  coordinate of the bounding box of the glyph, and  $\epsilon$  is the correction for the vertical aspect range, usually for an overshoot.

Table 9: Vertical aspects, anchor  $Y$  references, and clearances of Libertinus

|             | v. aspect | regular            | italic             | semibold | semibold italic    |
|-------------|-----------|--------------------|--------------------|----------|--------------------|
| <i>yra</i>  | ascender  | 688, [688, 698]    | 688, [688, 698]    | 690      | 698                |
| <i>ycr</i>  | capital   | 645, [645, 658]    | 645, [645, 658]    | 645      | 645                |
| <i>yrx</i>  | x-height  | 429, [429, 432]    | 429, [429, 432]    | 433      | 434, [433, 447]    |
| <i>yr0</i>  | baseline  | 0, [-12, 0]        | 0, [-12, 0]        | 0        | 0                  |
| <i>yrd</i>  | descender | -232, [-238, -227] | -233, [-238, -227] | -238     | -232, [-238, -232] |
| <i>ayra</i> | ascender  | 885                | 890                | 885      | 890                |
| <i>ayrc</i> | capital   | 850                | 850                | 805      | 850                |
| <i>ayrx</i> | x-height  | 645                | 645                | 645      | 645                |
| <i>byr0</i> | baseline  | -110               | -110               | -110     | -110               |
| <i>byrd</i> | descender | -319               | -319               | -319     | -319               |
| <i>ayda</i> | ascender  | 187                | 202                | 195      | 192                |
| <i>aydc</i> | capital   | 205                | 205                | 205      | 205                |
| <i>aydx</i> | x-height  | 216                | 216                | 212      | 198                |
| <i>byd0</i> | baseline  | 110                | 110                | 110      | 110                |
| <i>bydd</i> | descender | 87                 | 87                 | 81       | 100                |

## Base coverage

Libertinus serif regular has good above- and below-anchor base coverage for A–Z, a–z, and IPA letters, in the sense that most anchors are defined. But I will show later that some anchors are not set to good values. The base coverage gets worse for italic and semibold. Observe that many IPA letters reuse  $ax = 209$  and  $bx = 196$ . These are good anchors for the letter a, but these were just copied to some IPA letters without regard to the letterform and are therefore bad anchors for those IPA letters.

## Anchor survey, with patches

I will systematically survey above- and below anchors in the four font styles, commenting on any features and patching any problems. Refer to Tables 1 and 2 for anchor values. Missing anchors are readily apparent as blank entries. Large anchor-midpoint offset values ( $ax\delta$ ,  $bx\delta$ ) indicate possible problems, but these can often be explained by glyph symmetry.

### A–Z, regular, above- and below-anchor

The above-anchors of A–Z regular are good.

The above-anchor of D is just right of the major stem, not right enough to be centered over the bowl. For D,  $ax = 306$ . Consider the precomposed characters  $\text{D}^{\circ}$  (U+010E) and  $\text{D}^{\circ}$  (U+1E0A). The designer drew their glyphs with the mark centered at  $X = 312$  and  $296$  respectively. So, this is a deliberate design decision to keep the mark closer to the stem, not centered over the bowl.

Similarly, the above anchor of F is just right of the major stem, not right enough to be centered over cross strokes. For F,  $ax = 284$ . For  $\dot{\text{F}}$  (U+1E1E), the dot is centered at  $X = 254$ . So, the designer is deliberately choosing to keep the mark closer to the stem.

The above-anchor of L is centered over the major stem, which is a valid design choice.

The below-anchors of A–Z regular are good. But J's below anchor is too low, and Q is missing a below-anchor. Even though J and Q arguably do not need below-anchors for linguistic notation, I will set this anchor by the heuristic algorithm.

#### Patch J regular, below-anchor

J's below-anchor is too low. For J,  $by = 319$ , but J is not a full descender; its  $\text{bbox } Y_{\min} = -172$ , which is well above  $yr\text{d} = -232$ . Yes, the designer set  $by = byr\text{d} = -319$  for J in all four font styles, but this is arguably too low. According to the heuristic algorithm, the below-anchor anchor should be at  $by = Y_{\min} - by\delta d + \epsilon = -172 - 87 + 5 = -254$ , where  $\epsilon = 5$  accounts for the overshoot of J's tail.

#### Patch Q regular, below-anchor

Q has  $\text{bbox } Y_{\min} = -209$ . Again, according to the heuristic algorithm, the below-anchor anchor should be at  $by = Y_{\min} - by\delta d + \epsilon = -209 - 87 + 5 = -291$ , where  $\epsilon = 5$  accounts for the overshoot of Q's tail. For  $bx$ , observe that O and Q have the same counter outline, literally. So, the below-anchor for Q should be at  $bx = 350$ , the same as that for O.

### a–z, regular, above- and below-anchor

b, f, j, and t are missing above-anchors. Some letters (n, p, r, v, w, x, z) should have  $ay$  regularized to 645, not 646. For d, h, and k, the above-anchor is centered above the body, not the ascender; this is good design choice, but care must be taken to avoid collision between an above-mark and an ascender. For d and h,  $ay$  is at the x-height anchor reference, and therefore wide marks collide with the ascender. For k,  $ay$  is a bit higher, but not at the full ascender anchor references, and therefore the widest marks collide with the ascender.

The below-anchor of k is too low. Otherwise, the below-anchors of a–z are good.

#### Patch k regular, below-anchor

For k,  $by = 214$ . But it should be at  $byr0 = -110$ , like all other letters that sit on the baseline.

Here is this positioning of k in a patched version of the font (with o for comparison only).

Original:  $\text{k } \text{ko } \text{ok}$

Patched:  $\text{k } \text{ko } \text{ok}$

#### Patch b, d, f, h regular, above-anchor

For b, d, f, and h, set  $ay = ayr0 = 885$ .

For  $ax$  for b, d, f, and h, there are at least two strategies. First, set  $ax$  at the bowl apex. So, for f, set  $ax = 279$  at the apex of the bowl. But for b, d, and h, you could set  $ax$  centered horizontally over the bowl, in which case you also need to decide which two vertical aspects of the bowl should be used for the centering.

#### Patch j regular, above-anchor

j needs an above anchor for the rare case of an above-mark that is placed over the dot. i does have an above-anchor at (137, 795), and the outlines of the top of

### Patch n, p, r, v, w, x, z regular, above-anchor

In my patched version of the font, I regularize *ay* to *ayr0* = 645 for all letters that sit at the x-height, including n, p, r, v, w, x, z.

### Above-anchor, a–z regular

b, f, j, and t are missing the above-anchor. d and h have an above-anchor above the meanline but not above the ascender. l has an above-anchor above the ascender. k has an above-anchor, but it is just below the ascender, so any marks other than the do crash into the ascender. These are all legitimate design choices. There are no above-marks, other than those with precomposed characters, the would ever be combined with b, f, or t.

### Patch j dot removal for three above-marks

The dot of i and j should be removed before adding any above-mark. But the dot is not removed for three above-marks: x, turned tilde, double macron. So the GSUB table should be patched to fir the dotless j substitution in this context.

To do.

### Capital mark alternates

Libertinus has custom glyphs for some marks (acute, grave, circumflex, breve, a.o.) that are designed better for capital letters. These alternates are flatter. GSUB lookups 23 and 24 implement the substitution. This substitution is not fired on some capital consonants (T, V, W, X) because these marks are not expected on these bases.

Libertinus has custom glyphs for some marks that are designed better for superscripts. These alternates are smaller. Regrettably?, no GSUB table implements their substitution.

A circumflex Â Ã Ä Å Æ

O circumflex Ô Õ Ö Ø Ñ

AE ligature Æ Æ Æ Æ Æ

Š Ţ Ÿ Š Ţ Ÿ Š Ţ Ÿ

Š Ţ Ÿ Š Ţ Ÿ Š Ţ Ÿ

### Regular, above-anchor, capitals, *X* coordinate

Figure 3 (left panel) shows pairs of bases aligned at their above-anchor. If a capital letter has a bad *X* coordinate for its above-anchor, then it will look always off, right or left, in its black column relative to the gray letters that it overlays. Note that the above-anchor *X* coordinate is not in the geometric (horizontal) center of the glyph, by design. The above-anchor *X* coordinate is always off center by a few UPM units, sometimes left and sometimes right. But it very hard to discern a few UPM units. See the right panel of Figure 3 for an illustration of increasingly bad (spoofed) *X* coordinates of N's above-anchor.

Note that the above-anchor of L is above the major (vertical) stem of the L. For E, F, P and R, the above-anchor is optically centered above the glyph, not above the major stem.

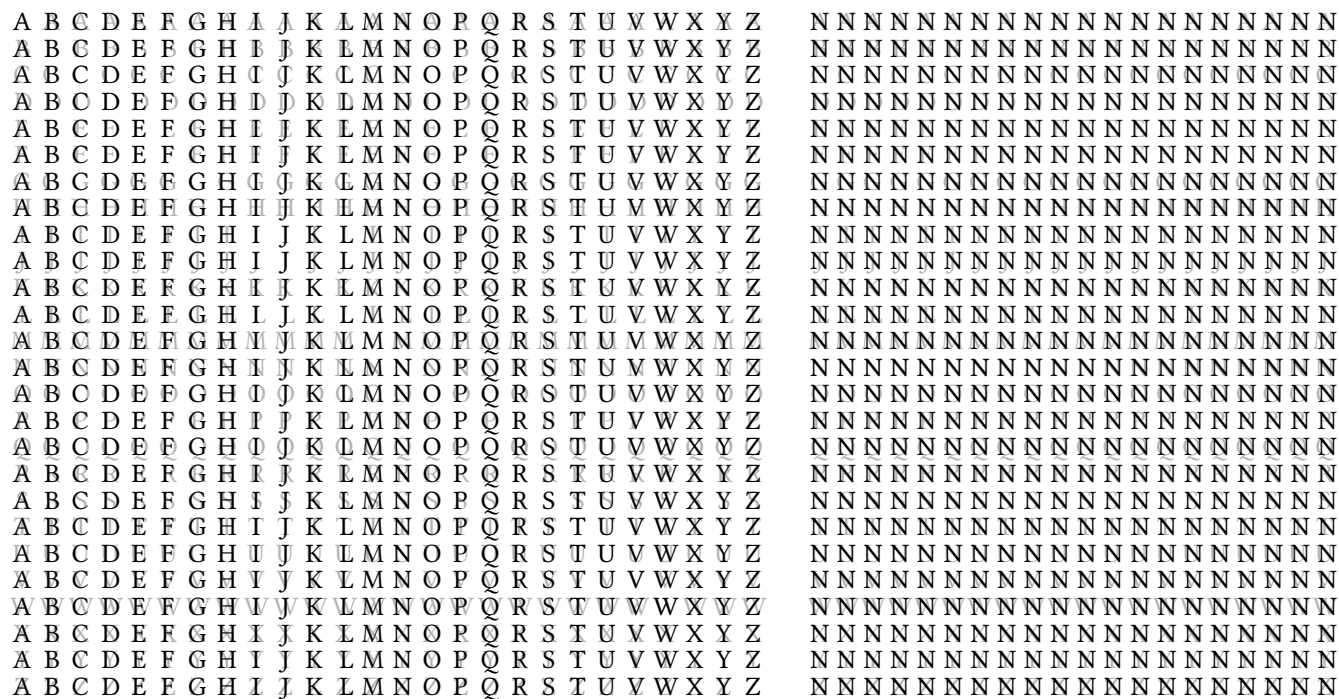


Figure 3: Above-anchor  $X$  coordinate alignment. In the left panel, the 26×26 grid shows letters A–Z aligned at their above-anchor. The right panel shows how hard it is to discern bad  $X$  coordinates. The right panel shows N aligned with letters at their above-anchor, starting at the first column with Ns above-anchor as currently set by the font, and then spoofing increasing bad values for the  $X$  coordinate of N's above-anchor, by 5 UPM per column. At the rightmost column, it becomes evident that the (spoofed)  $X$  coordinate of N's above-anchor is bad, because in that column the black Ns are too far right of the gray letters below them.

## Comma-style marks

Libertinus has some (precomposed) glyphs for Unicode characters (with code points) where the caron or cedilla has a comma-style shape, instead of a wedge or hook as might be naively expected. But all of these are appropriate for Czech/Slovak orthography (caron or *háček*) or Latvian orthography (cedilla, but really the comma-above mark or *komatiņš*, by a known mistake in Unicode).

### Czech/Slovak *háček*

In Czech/Slovak orthography, the d, l, L, n and t may take a reduced caron that may look like a comma, not a wedge. In Libertinus, only the d, L, l, and t have the reduced comma-style caron in their precomposed form, and the n has wedge caron precomposed form.

`\char"010F` **ď** *ď* **ď** *ď*

`\char"013D` **Ľ** *Ľ* **Ľ** *Ľ*

`\char"013E` **ĺ** *ĺ* **ĺ** *ĺ*

`\char"0148` **ň** *ň* **ň** *ň*

`\char"0165` **ť** *ť* **ť** *ť*

Libertinus even has a custom glyph for a small capital L with a caron, and it is comma-style.

`\textsc{1\char"030C}` **ɽ**

`\textsc{\char"013E}` **ɽ**

### Cedilla and Latvian *komatiņš*

Here are all precomposed characters with a cedilla.

`\char"00C7` **Ç** *Ç* **Ç** *Ç*    `\char"00E7` **ç** *ç* **ç** *ç*

`\char"1E08` **Ĉ** *Ĉ* **Ĉ** *Ĉ*    `\char"1E08` **ĉ** *ĉ* **ĉ** *ĉ*

`\char"1E10` **Ď** *Ď* **Ď** *Ď*    `\char"1E11` **ď** *ď* **ď** *ď*

`\char"0228` **Ē** *Ē* **Ē** *Ē*    `\char"0229` **ē** *ē* **ē** *ē*

`\char"1E1C` **Ě** *Ě* **Ě** *Ě*    `\char"1E1D` **ě** *ě* **ě** *ě*

`\char"0122` **Ĝ** *Ĝ* **Ĝ** *Ĝ*    `\char"0123` **ĝ** *ĝ* **ĝ** *ĝ*

`\char"1E28` **Ĥ** *Ĥ* **Ĥ** *Ĥ*    `\char"1E29` **ĥ** *ĥ* **ĥ** *ĥ*

`\char"0136` **Ķ** *Ķ* **Ķ** *Ķ*    `\char"0137` **ķ** *ķ* **ķ** *ķ*

`\char"013B` **Ļ** *Ļ* **Ļ** *Ļ*    `\char"013C` **ļ** *ļ* **ļ** *ļ*

`\char"0145` **Ņ** *Ņ* **Ņ** *Ņ*    `\char"0146` **ņ** *ņ* **ņ** *ņ*

`\char"0156` **Ŗ** *Ŗ* **Ŗ** *Ŗ*    `\char"0157` **ŗ** *ŗ* **ŗ** *ŗ*

`\char"015E` **Ș** *Ș* **Ș** *Ș*    `\char"015F` **ș** *ș* **ș** *ș*

`\char"0162` **Ț** *Ț* **Ț** *Ț*    `\char"0163` **ț** *ț* **ț** *ț*

These characters – Ģ ģ Ķ ķ Ļ ļ Ņ ņ Ŗ ŗ – use the Latvian *komatiņš* which is proper for Latvian orthography, but the result of a historical mistake in Unicode naming.

The Unicode characters of D and d with cedilla exist for scholarly and historical reasons, not Latvian orthography. Fonts design these Unicode characters differently. Libertinus designs them inconsistently: comma-style in regular and italic, and hook-style in (semi)bold and (semi)bold italic. I think the cedilla on a D or d should be hook-like, but I do not really know.

The other characters – c, e, h, s, t – with a cedilla should definitely be hook-like, and they are so in Libertinus.

Somewhat mysteriously, if the LaTeX source tries to combine any of these letters – c, d, e, g, h, k, l, n, r, s, t – with the combining cedilla, U+0327, then the precomposed character is produced.

`c\char"0327` ċ ċ ċ  
`d\char"0327` ḋ ḋ ḋ  
`e\char"0327` ě ě ě  
`g\char"0327` ġ ġ ġ  
`h\char"0327` ĥ ĥ ĥ  
`k\char"0327` ķ ķ ķ  
`l\char"0327` ĺ ĺ ĺ  
`n\char"0327` ñ ñ ñ  
`r\char"0327` ŕ ŕ ŕ  
`s\char"0327` š š š  
`t\char"0327` ț ț ț

This behavior does not make sense for the Latvian *komatiņš* bases – g, k, l, n, r – because any person trying to produce Latvian g with *komatiņš* would not try `g\char"0327` or `\c{g}`. This behavior is tolerable for the other bases – c, d, e, h, s, t.

## IJ

The ij and IJ ligatures are Unicode characters (with code points) and are supported by Libertinus.

`IJ \char"0132` U+0132 capital ligature IJ

`ij \char"0133` U+0133 small ligature ij

Libertinus has custom glyphs for these ligatures with an acute, and the small capital ligature with an acute. The GSUB table that activates these custom glyphs only works when the language is set to Dutch by a font loading command. For example,

`\newfontfamily\SerifRegularDutch[Language = Dutch]{LibertinusSerif-Regular.otf}`

`İf {\SerifRegularDutch \char"0132\char"0301}` Dutch capital ligature IJ with acute

`íj {\SerifRegularDutch \char"0133\char"0301}` Dutch small ligature ij with acute

`İf {\SerifRegularDutch \textsc{\char"0133\char"0301}}` Dutch small capital ligature IJ with acute

In an English LaTeX document, ij and IJ are kerned by the GPOS table to produce ij and IJ, so they look close together, but they are not substituted by the single glyph for the ligature.

## Tilde below, double acute above, double grave above, on vowels, in italic and semibold italic

Consider the tilde below. There is the combining tilde below, U+0330, encoded in LaTeX as the codepoint `\char"0330`. Remember to put the base before the combining mark in LaTeX, e.g. `a\char"0330` ã. Three vowels (e, i, u) have precomposed characters – lowercase and capital – encoded in LaTeX as these codepoints:

`\char"1E1A \char"1E1B \char"1E2C \char"1E2D \char"1E74 \char"1E75` Ē ē Ĭ ĭ Ū ū Ē ē Ĭ ĭ Ū ū Ē ē Ĭ ĭ Ū ū

E, e, I, i, U, or u followed by `\char"0330` will produce the precomposed character glyph. A, a, O, and o, as well as any other IPA vowel with a below-anchor, followed by `\char"0330` will be shaped by attaching the combining below tilde to that vowel at their below-anchor. Note the specific case of i. `i\char"0330` = `\char"1E2D` ï = ï is a dotted i (a precomposed glyph).

Suppose you want a vowel with a tilde below, possibly also with a mark above, especially acute, grave, caron, double acute, or double grave. In the specific case of i, the above mark will crash into the dotted i unless you take care to encode the triplet starting with a dotless i, U+0131. For example, consider the combining acute U+0301.

`i\char"0330\char"0301` = `\char"1E2D\char"0301` ï̃ ï̃ ï̃ = ï̃ ï̃ ï̃

`\char"0131\char"0330\char"0301` ï̃ ï̃ ï̃



Recall that the canonical order for a triplet of a base and two combining marks (one below and one above) is base + combining below mark + combining above mark, not base + above + below.

Consider the wrong order:

```
i\char"0301\char"0330 i i i i\char"0301 i i i i\char"0330 i i i i
\char"00EC\char"0330 i i i i\char"00EC i i i i
\char"1E2D\char"0301 i i i i\char"1E2D i i i i
```

But note that the position of the combining tilde below is bad in italic and semibold italic. This is because the below-anchor for the combining tilde below mark is bad. For tilde below, the precomposed character glyphs with i, e, and u are sometimes good and sometimes bad in italic and semibold italic.

Precomposed characters are blue and therefore cannot be changed unless their glyph outline is changed. Black combinations of vowels and marks are shaped by anchors and therefore can be changed by setting good anchors.

Use the combining grapheme joiner (CGJ) character U+034F, \char"034F, or \cgj if loaded by some package, to force the shaper to use combining marks instead of precomposed glyphs.

```
AEIOUEɔaeiouεɔ AEIOUEɔaeiouεɔ AEIOUEɔaeiouεɔ AEIOUEɔaeiouεɔ
ÁÉÍÓÚÊɔáéíóúés ÁÉÍÓÚÊɔáéíóúés ÁÉÍÓÚÊɔáéíóúés ÁÉÍÓÚÊɔáéíóúés
ÀÈÌÒÛÈɔàèìòùèɔ ÀÈÌÒÛÈɔàèìòùèɔ ÀÈÌÒÛÈɔàèìòùèɔ ÀÈÌÒÛÈɔàèìòùèɔ
ĂĖİŎŬĖɔăėıŏŭěɔ ĂĖİŎŬĖɔăėıŏŭěɔ ĂĖİŎŬĖɔăėıŏŭěɔ ĂĖİŎŬĖɔăėıŏŭěɔ
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```

Patched

```
AEIOUEɔaeiouεɔ AEIOUEɔaeiouεɔ AEIOUEɔaeiouεɔ AEIOUEɔaeiouεɔ
ÁÉÍÓÚÊɔáéíóúés ÁÉÍÓÚÊɔáéíóúés ÁÉÍÓÚÊɔáéíóúés ÁÉÍÓÚÊɔáéíóúés
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ÀÈÌÒÛÈɔàèìòùèɔ ÀÈÌÒÛÈɔàèìòùèɔ ÀÈÌÒÛÈɔàèìòùèɔ ÀÈÌÒÛÈɔàèìòùèɔ
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ĂĖİŎŬĖɔăėıŏŭěɔ AEİŎŬĖɔăėıŏŭěɔ AEİŎŬĖɔăėıŏŭěɔ AEİŎŬĖɔăėıŏŭěɔ
```

## Mark over space

I added a new en-width x-height PUA character, U+E100, with anchors. To float an above mark, say double grave U+030F over a space, write `\char"E100\char"030F` But in practice, the floating mark carries the same tonal specification as the preceding vowel, and therefore must align exactly with the height of that vowel's combining mark. To ensure this mark alignment, insert a combining grapheme joiner, `\cgj`, in the prior vowel + mark combo, to force anchor positioning without substituting a precomposed glyph there. Note that in Libertinus, the two graves in the U+030F glyph are not translates of an identical outline; this is a deliberate design choice by the designer.

Original

`-ke\char"030F\ \char"030F -kë" -kë" -kë" -kë"`

Patched without CGJ

`-ke\char"030F \char"E100\char"030F -kë" -kë" -kë" -kë"`

Patched with CGJ

`-ke\cgj\char"030F \char"E100\char"030F -kë" -kë" -kë" -kë"`

## f with dot below

tiyó ṛfa

*tiyó ṛfa*

**tiyó ṛfa**

***tiyó ṛfa***

`ṛ = \char"1E5B` is the precomposed character U+1E5B Latin small letter r with dot below.

`ḟ = f\char"0323` is f with U+0323 combining dot below.

Here are some letters with a dot below.

`a b d e h i k l m n o r s t u v w y z c f g j p q x`

*a b d e h i k l m n o r s t u v w y z c f g j p q x*

**a b d e h i k l m n o r s t u v w y z c f g j p q x**

***a b d e h i k l m n o r s t u v w y z c f g j p q x***

The left group of letters – `a b d e h i k l m n o r s t u v w y z` – have a precomposed character. For the right group of letters – `c f g j p q x` – the dot is placed according to the below-anchor (if the below-anchor is defined for the letter) or by HarfBuzz fallback shaping (placing the dot below the baseline, regardless of the letter's bounding box).

In all styles – regular, italic, semibold, and semibold italic – the precomposed glyph of y with dot below places the dot beside the descender, even though the y has a below-anchor below the descender.

Do the letters c, f, g, j, p, q, and x have good below-anchors? In regular, the below-anchor of all these combining bases is below the glyph outline (good). In italic, the below-anchor of f, p, and q is beside the descender (bad), the below-anchor of g and j is below the descender (good), and the below-anchor of c and x is below the baseline (good). In semibold, the below-anchor of g and j is below the descender (good), but the other bases – c, f, p, q – have no below-anchor, and HarfBuzz fallback positioning places the combining dot below the baseline, not below the bounding box; for c, f and x this is good, but for p and q this is bad, because of the descenders. In semibold italic, the below-anchor of c, g, j, p and q is below the glyph outline (good), the below-anchor of f is inside the descender (very bad), and x has no below-anchor but HarfBuzz fallback positioning places the combining dot below the baseline (good).

Which dots are used in precomposed characters and how big are the dots? In all styles (except semibold italic), the precomposed characters reference the dot of `uni0307`, combining dot above, U+0307, and take care to translate this above-mark to a position below the letter outline, except for m, which references `dotbelowcomb`, combining dot below, U+0323. In semibold italic, `dotaccent` is used instead of `uni0307`. These two dots are four-point Bézier circles (in regular and semibold, but ovals in italic and semibold italic), but of different diameter. In regular, `uni0307` has diameter 100, and `dotbelowcomb` has diameter 108. That is, the precomposed characters (except m) use a smaller dot, and the non-precomposed letters (and m) take

a bigger dot. In semibold, the size difference is reversed: uni0307 has larger diameter 124, and dotbelowcomb has small diameter 108. Actually, semibold dotbelowcomb is not a Bézier circle – the horizontal span is 109, and the vertical span is 108. In semibold italic, dotaccent is a skewed four-point Bézier oval – the horizontal and vertical spans are 120, offset by  $(\pm 6, \pm 7.5)$  from the center – and dotbelowcomb is slightly larger, slightly flatter skewed six-point Bézier oval – horizontal span is 122 and the vertical span is 120. So, in semibold italic the two dots in precomposed and combining form are closest in size. Recall that f has a bad below-anchor, inside the descender, but g, j, p and q have a good below-anchor, below the glyph outline.

So, in order to solve the problem of the bad dot below f in italic, I need to decide the scope of the badness to correct. The simplest patch is move the below-anchor in italic and semibold italic below the descender. But this does not solve inconsistent below dot size across styles and letters. In italic and semibold italic, g and j have a (good) below-anchor at  $Y = -319$ , so that is also good Y coordinate for f's below-anchor.

I asked Copilot/GPT-5.1 to calculate a good X coordinate for f's below-anchor in italic and semibold italic, taking into account the U-pocket of f's descender and the below-anchor of dotbelowcomb. I decided that this U-pocket strategy was appropriate after testing other strategies that accounted for the slant of onf the major stroke. Here is a reasonably faithful summary of the calculations and reasoning by Copilot/GPT-5.1.

For italic, the descender's U-pocket is defined by the control points at (14, -238) and (33, -205), giving a U-center around  $x \approx 24$ . Using the (symmetric) italic dotbelowcomb anchor as the dot's optical center, the dot should sit under this U-pocket, not under the ball or the stem. A balanced, U-centered, slightly inner-biased choice is  $X \approx 30$ .

For semibold italic, the U-pocket of the semibold italic descender is wider and more asymmetric, with a geometric U-center of  $X = -33.5$ . The semibold italic dotbelowcomb itself is also skewed, with its anchor already shifted relative to its mass. In practice, the visually correct position for the dot below f is under the open side of the U-pocket, clearly right of the ball and closer the right of the stem. The anchor candidate is  $X \approx 62$ .

Here are examples with the patched fonts, with f's below-anchor in italic at (30, -319) and semibold italic at (62, -319). This does not solve the problem of inconsistent dot size below letters, and I did not touch p, q, or y.

Original    Patched italics (f only)

tiyô rfa    tiyô rfa

tiyô rfa    tiyô rfa

tiyô rfa    tiyô rfa

tiyô rfa    tiyô rfa

Original (repeated from above)

a b d e h i k l m n o r s t u v w y z c f g j p q x

a b d e h i k l m n o r s t u v w y z c f g j p q x

a b d e h i k l m n o r s t u v w y z c f g j p q x

a b d e h i k l m n o r s t u v w y z c f g j p q x

Patched italics (f only)

a b d e h i k l m n o r s t u v w y z c f g j p q x

a b d e h i k l m n o r s t u v w y z c f g j p q x



## Marks

Here is a catalog of combining marks in Unicode, grouped by the seven GPOS anchor class indices in Libertinus (0–6).

Above marks – GPOS anchor 0

[illegible]

**Below marks – GPOS anchor 2**

X X X X X X X X X X X X X X X X X X X X X X X X X X X X  
X X X X X X X X X X X X X X X X X X X X X X X X X X X X  
X X X X X X X X X X X X X X X X X X X X X X X X X X X X  
X X X X X X X X X X X X X X X X X X X X X X X X X X X X

Left angle – GPOS anchor 3

$$\mathbf{X}^T \mathbf{x}^T \bar{\mathbf{X}} \bar{\mathbf{x}}$$

Below-right marks – GPOS anchor 4

$$\begin{array}{c} \mathbf{x}_j \mathbf{x}_l \mathbf{x}_z \\ \mathbf{x}_j \mathbf{x}_l \mathbf{x}_z \\ \mathbf{x}_j \mathbf{x}_l \mathbf{x}_z \\ \mathbf{x}_j \mathbf{x}_l \mathbf{x}_z \end{array}$$

## Cedilla – GPOS anchor 5

$$\mathbf{X}_3 \quad \mathbf{x}_3 \quad \mathbf{X}_3 \quad \mathbf{x}_3$$

### Overlay marks – GPOS anchor 6

$\emptyset \emptyset \emptyset \emptyset \emptyset$   
 $\emptyset \emptyset \emptyset \emptyset \emptyset$   
 $\emptyset \square \emptyset \square \emptyset \square \emptyset \square \emptyset$   
 $\emptyset \square \emptyset \square \emptyset \square \emptyset \square \emptyset$

Double marks

$\overline{xx} \overline{xx} \overline{xx} \overline{xx} \overline{xx} \overline{xx}$   
 $\widetilde{xx} \widetilde{xx} \widetilde{xx} \widetilde{xx} \widetilde{xx} \widetilde{xx}$   
 $\widehat{xx} \widehat{xx} \widehat{xx} \widehat{xx} \widehat{xx} \widehat{xx}$   
 $\check{xx} \check{xx} \check{xx} \check{xx} \check{xx} \check{xx}$   
 $x\grave{x} x\grave{x} x\grave{x} x\grave{x} x\grave{x} x\grave{x}$

Greek marks

$\acute{\tau} \acute{\tau} \acute{\tau} \acute{\tau} \acute{\tau} \acute{\tau}$   
 $\grave{\tau} \grave{\tau} \grave{\tau}$   
 $\tilde{\tau} \tilde{\tau} \tilde{\tau} \tilde{\tau} \tilde{\tau} \tilde{\tau}$   
 $\bar{\tau} \bar{\tau} \bar{\tau} \bar{\tau} \bar{\tau} \bar{\tau}$   
 $\acute{\tau} \acute{\tau} \acute{\tau}$   
 $\grave{\tau} \grave{\tau} \grave{\tau} \grave{\tau} \grave{\tau} \grave{\tau}$   
 $\tilde{\tau} \tilde{\tau} \tilde{\tau} \tilde{\tau} \tilde{\tau}$   
 $\acute{\tau} \acute{\tau} \tau$   
 $\acute{\tau} \acute{\tau} \acute{\tau} \acute{\tau} \acute{\tau} \acute{\tau}$   
 $\tilde{\tau} \tilde{\tau} \tilde{\tau} \tilde{\tau} \tilde{\tau}$   
 $\acute{\tau} \acute{\tau} \tau$   
 $\acute{\tau} \acute{\tau} \acute{\tau} \acute{\tau} \acute{\tau} \acute{\tau}$



Above-right marks – GPOS anchor 1

Libertinus has three combining marks with GPOS anchor 1 – U+031B horn, U+0315 comma above right, and U+0358 dot above right.

A few bases in Libertinus have this anchor (anchor 1), but none that possibly need it.

## Horn

The Vietnamese alphabet has two horn vowels – Ō ơ Ūư – and these vowels may take tonal marks. All these combinations are Unicode characters (with code points).

ŌōŪūŌóŌòŌỏŌõŌôŪúŪùŪủŪũŪu

O o U u Ö ö Õ õ Ö ö Õ õ O o Ú ú Û û Ü ü Û û U u

O o U u Ó ó Ò ò Ô ô Õ õ Ö ö Ú ú Û û Ü ü

O o U u Ó ó Ò ò Ô ô Õ õ Ö ö Ø ø Ù ú Û û Ü ü U u

There is a Unicode character for the combining horn mark: `\char"031B U+031B`, but this is not used in practice. The Vietnamese vowels are usually entered in a LaTeX document as Unicode characters, also with `\usepackage[vietnamese]{babel}` for hyphenation and spacing.

Comma above right

There is a Unicode character for the combining comma above right mark: `'\char"0315 U+0315`.

U+0315 (comma above right) is used in rare cases when another comma-like mark does not fit.

In IPA, the glottalization diacritic mark uses U+0313 (comma above) as a combining mark. But when a glottalization mark is needed for a superscript consonant (1) or superscript consonant cluster (2), then the combining mark U+0315 (comma above right) is used instead, because its positioning is hopefully better than that of U+0313.

(1) *kʷkt* [*kʷkʰt*] ‘to scrape’ (Nuxalk)

(2) *p'ak* [p<sup>kx</sup>ak] 'he/she cut it' (Upper Chehalis)

In (1), the superscript consonant cluster is <sup>wkx</sup> and it is glottalized. In (2), the superscript consonant <sup>x</sup> is glottalized.

Because U+0315 is being used in superscript position on a smaller sized glyph, U+0315 should probably be smaller than U+0313, but it is not so in Libertinus.

Dot above right

There is a Unicode character for the combining dot above right mark: `\char"0358 U+0358 dot above right`

U+0358 should be a small, above-right dot, used to mark palatalization or similar secondary articulations on superscript consonants, created for the same typographic reason U+0315 exists – the standard palatalization diacritic (U+0307, dot above) is too big and too high.

In Libertinus, U+0358 is a 104-diameter Bezier circle, and U+0307 is a smaller Bezier oval, inverting the desired size difference.

There are only a few IPA superscript letters that may serve as the possible host of U+0315 or U+0358 on superscript consonants or on superscript consonant clusters. In Libertinus, regrettably none of these characters has the anchor (anchor 1), so the mark (U+0315 and U+0358) is positioned by HarfBuzz fallback shaping, which seems to place the mark at the right boundary of the glyph at a constant vertical aspect (effectively at the superscript x-height).

These IPA marks U+0315 and U+0358 should only be used in regular (not italic and not semibold). Regrettably, in Libertinus serif semibold, U+0358 looks like a grave, not a dot, and there is no U+0358 glyph in Libertinus serif semibold italic.

There are three comma/apostrophe glyphs that could possibly mark glottalization: U+0313 combining comma above, U+0315 combining comma above right, and U+02BC modifier letter apostrophe. In Libertinus, U+0315 has the worst appearance, even though U+0315 is intended for superscript letters. In Gentium Plus (SIL Global), U+0315 is tightly above-right the superscript letter, as expected.



Libertinus

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t] [k<sup>wkx̣</sup>t]

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t] [k<sup>wkx̣</sup>t]

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t] [k<sup>wkx̣</sup>t]

Gentium Plus

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0313 t] [k<sup>wkx̣</sup>t]

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t] [k<sup>wkx̣</sup>t]

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"02BC t] [k<sup>wkx̣</sup>t]

I patched Libertinus to add anchors to those IPA superscript letters to support U+0315.

Patched Libertinus, selected IPA superscript letters with U+0315 and U+0358

ḥḥj̣ṛɹ̣ɹ̣ɹ̣ɹ̣ẉỵỵḷṣx̣ḅḳṃp̣ṭụṿ.ɹ̣β̣ɣ̣δ̣

ḥḥj̣ṛɹ̣ɹ̣ɹ̣ẉỵỵḷṣx̣ḅḳṃp̣ṭụṿ.ɹ̣β̣ɣ̣δ̣

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t] [k<sup>wkx̣</sup>t]

Gentium Plus, selected IPA superscript letters with U+0315 and U+0358

ḥḥj̣ṛɹ̣ɹ̣ɹ̣ẉỵỵḷṣx̣ḅḳṃp̣ṭụṿ.ɹ̣β̣ɣ̣δ̣

ḥḥj̣ṛɹ̣ɹ̣ɹ̣ẉỵỵḷṣx̣ḅḳṃp̣ṭụṿ.ɹ̣β̣ɣ̣δ̣

[k\char"0313\char"02B7\char"1D4F\char"02E3\char"0315 t] [k<sup>wkx̣</sup>t]

Here are the IPA superscript letters considered and their new anchor (anchor 1).

U+02B0: (347, 648), U+02B1: (339, 648), U+02B2: (193, 648), U+02B3: (284, 648), U+02B4: (274, 648), U+02B5: (319, 648), U+02B6: (308, 648), U+02B7: (497, 648), U+02B8: (368, 648), U+02E0: (341, 648), U+02E1: (193, 648), U+02E2: (270, 648), U+02E3: (344, 648), U+02E4: (282, 648), U+1D47: (319, 648), U+1D4F: (353, 648), U+1D50: (533, 648), U+1D56: (334, 648), U+1D57: (241, 648), U+1D5A: (525, 648), U+1D5B: (354, 648), U+1D5C: (508, 648), U+1D5D: (335, 648), U+1D5E: (372, 648), U+1D5F: (353, 648)

I did not change the anchor-1 of U+315 and U+0358. U+0358 has its anchor at the xMin and yMin of its bounding box. But U+0315 already has some clearance built in to its anchor: its anchor is 62 units below its yMin, but 25 units right of it xMin. So, I decided to provide x-clearance of 12 units and y-clearance of 80 units from the bases. All these IPA superscript letters already have a superscript meanline at  $Y = 630$ . So their anchor will be at  $Y = 630 - 62 + 80 = 648$ . The X coordinate depends on their bounding box and its xMax. To provide the desired clearance of 12 units, the anchor is at  $X = \text{xMax} + 26 + 12$ . These anchor coordinate values were optimized for U+0315 but they work fine for U+0358.





No marks: rare, digraph etc.

Ƨ ɤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj  
Ƨ ɤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj  
Ƨ ɤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj  
Ƨ ɤ DZ Dz dz DŽ Dž dž LJ Lj lj NJ Nj nj

Small capitals

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Đ Ć Þ Ĳ ß ŋ  
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Đ Ć Þ Ĳ ß ŋ  
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Đ Ć Þ Ĳ ß ŋ  
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Æ Đ Ć Þ Ĳ ß ŋ

Precomposed characters of anchor-relevant bases and marks

à á â ã ä å æ ç è é ê ë ì í î ï ð ó ô õ ö ø ò ò ò ù ú û ü ÿ ý ÿ ÿ x d l n r s t z  
à á â ã ä å æ ç è é ê ë ì í î ï ð ó ô õ ö ø ò ò ò ù ú û ü ÿ ý ÿ ÿ x d l n r s t z  
à á â ã ä å æ ç è é ê ë ì í î ï ð ó ô õ ö ø ò ò ò ù ú û ü ÿ ý ÿ ÿ x d l n r s t z  
à á â ã ä å æ ç è é ê ë ì í î ï ð ó ô õ ö ø ò ò ò ù ú û ü ÿ ý ÿ ÿ x d l n r s t z

## Base and mark shaping – Linguistics notation

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic.

The grids use Latin in IPA bases and combining marks that are used in linguistics notation.

A black color means GPOS anchor positioning.

A **blue** color means a precomposed glyph.

A **red** color means HarfBuzz fallback shaping (no anchors, no precomposed glyph, no GSUB substitution).

A faded color – gray, light blue, or pink – means that the base and mark combination is not reasonably expected in notation in linguistics.

A gray color can also mean that mark glyph is missing in the font. In this case, just the gray base is printed. This only happens in Libertinus Serif Semibold and Semibold Italic for three below marks: ◌̑ (U+0333 double macron below), ◌̐ (U+033A inverted bridge below), and ◌̑ (U+033B square below). Otherwise, all considered base and mark combinations can be drawn in XeLaTeX documents in Libertinus without tofu for missing glyphs.

## Regular, Bases requiring anchors for common marks, Common combining marks

[illegible]

## Regular patched, Bases requiring anchors for common marks, Common combining marks

[illegible]









## Semibold patched, Bases requiring anchors for common marks, Common combining marks

[illegible]

## Semibold italic, Bases requiring anchors for common marks, Common combining marks

à B c d è f g i j k l m n ò p q r s t u v w x y z æ œ ñ è à ò B d d é é é j g g è Y y q h l l l k w m j n ñ N ò é i j l r r s s f u A A Y z z z j l  
á B c d è f g i j k l m n ó p q r s t u v w x y z æ œ ñ é á ó B d d é é é j g g è Y y q h l l l k w m j n ñ N ó é i j l r r s s f u A A Y z z z j l  
â B c d è f g i j k l m n ô p q r s t u v w x y z æ œ ñ ê â ô B d d é é é j g g è Y y q h l l l k w m j n ñ N ô é i j l r r s s f u A A Y z z z j l  
ã B c d è f g i j k l m n õ p q r s t u v w x y z æ œ ñ ã õ B d d é é é j g g è Y y q h l l l k w m j n ñ N õ é i j l r r s s f u A A Y z z z j l  
ā B c d ē f g i j k l m n ō p q r s t ū v w x y z æ œ ñ ē ā ō B d d é é é j g g è Y y q h l l l k w m j n ñ N ē ā ō é i j l r r s s f u A A Y z z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è Y y q h l l l k w m j n ñ N é a o é i j l r r s s f u A A Y z z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è Y y q h l l l k w m j n ñ N é a o é i j l r r s s f u A A Y z z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è Y y q h l l l k w m j n ñ N é a o é i j l r r s s f u A A Y z z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è Y y q h l l l k w m j n ñ N é a o é i j l r r s s f u A A Y z z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è Y y q h l l l k w m j n ñ N é a o é i j l r r s s f u A A Y z z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è Y y q h l l l k w m j n ñ N é a o é i j l r r s s f u A A Y z z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è Y y q h l l l k w m j n ñ N é a o é i j l r r s s f u A A Y z z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è Y y q h l l l k w m j n ñ N é a o é i j l r r s s f u A A Y z z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è Y y q h l l l k w m j n ñ N é a o é i j l r r s s f u A A Y z z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è Y y q h l l l k w m j n ñ N é a o é i j l r r s s f u A A Y z z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è Y y q h l l l k w m j n ñ N é a o é i j l r r s s f u A A Y z z z j l

## Semibold italic patched, Bases requiring anchors for common marks, Common combining marks

à B c d e f g i j k l m n ò p q r s t u v w x y z æ œ ñ è à ò b d d é é é j g g è y r q h l l l k u u u n ò è i j l r r s s f u à à y z z j l  
á B c d e f g i j k l m n ó p q r s t u v w x y z æ œ ñ é á ó b d d é é é j g g è y r q h l l l k u u u n ó è i j l r r s s f u á á y z z j l  
â B c d e f g i j k l m n ô p q r s t u v w x y z æ œ ñ ê â ò b d d é é é j g g è y r q h l l l k u u u n ô è i j l r r s s f u â â y z z j l  
ã B c d e f g i j k l m n õ p q r s t u v w x y z æ œ ñ ë ã ò b d d é é é j g g è y r q h l l l k u u u n õ è i j l r r s s f u ã ã y z z j l  
ā B c d e f g i j k l m n ō p q r s t u v w x y z æ œ ñ ē ā ō b d d é é é j g g è y r q h l l l k u u u n ō è i j l r r s s f u ā ā y z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y r q h l l l k u u u n o è i j l r r s s f u a a y z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y r q h l l l k u u u n o è i j l r r s s f u a a y z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y r q h l l l k u u u n o è i j l r r s s f u a a y z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y r q h l l l k u u u n o è i j l r r s s f u a a y z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y r q h l l l k u u u n o è i j l r r s s f u a a y z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y r q h l l l k u u u n o è i j l r r s s f u a a y z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y r q h l l l k u u u n o è i j l r r s s f u a a y z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y r q h l l l k u u u n o è i j l r r s s f u a a y z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y r q h l l l k u u u n o è i j l r r s s f u a a y z z j l  
a b c d e f g i j k l m n o p q r s t u v w x y z æ œ ñ e a o b d d é é é j g g è y r q h l l l k u u u n o è i j l r r s s f u a a y z z j l





[illegible]

[illegible]



[illegible]



[illegible]

[illegible]





## Base and mark shaping – Complete

These grids display base and mark rendering in XeLaTeX documents in Libertinus serif regular, italic, semibold and semibold italic. The grids use more complete sets of Latin and IPA bases and above and below marks.

**Blue** indicates that XeLaTeX uses a precomposed Unicode character glyph.

Black indicates that the mark is positioned using GPOS anchors.

**Red** indicates HarfBuzz fallback shaping: no precomposed glyph and no GPOS anchors are used.

**Light gray** indicates that either the base or the mark glyph is missing from the font. In this case only the base glyph is shown.







**Italic, Latin, Above**

[illegible]









[illegible]



## Regular, Latin, Below

[illegible]



Regular patched, Latin, Below

[illegible]



**Italic, Latin, Below**

[illegible]

Italic patched, Latin, Below

[illegible]

## Semibold, Latin, Below

[illegible]



## Semibold patched, Latin, Below

[illegible]









[illegible]



[illegible]

Italic patched, IPA, Above

[illegible]

[illegible]



[illegible]

[illegible]

[illegible]



Regular, IPA, Below

[illegible]



[illegible]

**Italic, IPA, Below**

[illegible]

[illegible]



[illegible]

[illegible]

[illegible]



[illegible]

## Capital alternates

These grids display capitals with alternate marks.

[illegible]

[illegible]

[illegible]

[illegible]



[illegible]

[illegible]

[illegible]

[illegible]